



FINAL YEAR PROJECT

Final Evaluation Report

SECUREWATCH

Virtual Cybersecurity learning and implementation application with user-friendly interface.

FYP Team

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Name of the Project:

F22-082-D-SecureWatch

Abstract

The use of the internet has increased and is increasing rapidly with every passing day. From socializing to paying your bills everything is now done using the internet. When all the important procedures are carried out on the internet, all the malicious activities have also shifted to the internet as well. Considering the current situation, our society is the most vulnerable to these malicious intents as we lack knowledge about these attacks. Millions of people fall prey to these attacks daily. Social engineering, XSS, CSRF, Phishing, malware, and SQL injection attacks are being carried out daily. The solution to this issue is to make people aware of the rising attacks on the internet. They must know how to safely use the internet and if they encounter such an issue what could be the possible action in that situation?

A cybersecurity untrained employee can significantly threaten the organization's reputation, data, and integrity. Without training and proper knowledge, the employee can intentionally or unintentionally engage in activities that expose the organization to vulnerabilities. Inadequate awareness about password security, password protective measures, and secure password practices can create such easy vulnerabilities for exploitation. That makes investing in cybersecurity an essential and crucial part of our internet usage. These can help in the mitigation of attacks and maintaining a strong defense against the world of attacks.

Keeping in view the sensitivity of the issue we have proposed a system that provides the user with knowledge of the current malicious activities. Not only that the system evaluates the knowledge of the user. Since the knowledge is insufficient, we have developed lab simulations in a protected environment that allows users to test and practice their knowledge. Also lead them towards mitigation techniques to help them stay safe in the future.

Contents

Contents

FYP Team.	2
Members.....	2
Supervisors: Supervisor:	2
Name of the Project:	2
Abstract.....	3
Contents.....	4
1. Introduction to the Project:	7
2. Project vision:	8
2.1. Problem Statement.....	8
2.2. Business Opportunity.....	8
2.3. Objectives.....	8
2.4. Project Scope.....	8
2.5. Constraints.....	9
2.6. Stakeholder Description.....	10
2.6.1. Stakeholder Summary:	10
2.6.2. Goals and Problems of Stakeholders:	10
3. Software Requirement Specifications:	10
3.1. List of Features.....	10
3.2. Functional Requirements:	11
3.3. Quality Attributes:	12
3.4. Non-Functional Requirements:	12
4. High-level Use Cases:	14
4.1. Use Case Diagram.....	22
5. Iteration Plan.....	23
6. Iteration 1.....	24
6.1. Use cases covered.....	24

Sunday, 4th June, 2023.

6.1.1. Extended Use Cases.....	24
6.1.2. System Sequence Diagrams.....	27
6.1.3. Activity Diagrams.....	30
6.2. Architecture Diagram.....	33
6.3. Domain Model Diagram.....	34
6.4. Sequence Diagrams.....	35
6.4.1. Scenario 1:	35
6.4.2. Scenario 2:	36
6.4.3. Scenario 3:	37
6.4.4. Scenario 4:	38
6.4.5. Scenario 5:	38
6.4.6. Scenario 6:	39
7. Implementation Details for Iteration 01.....	41
8. Iteration 2.....	42
8.1. Use cases covered.....	42
8.1.1. Extended Use Cases.....	42
8.1.2. System Sequence Diagrams.....	45
8.1.3. Activity Diagrams.....	47
8.1.4. Domain Model Diagram.....	49
8.3. Deployment Diagram.....	51
9. Implementation Details for Iteration 02.....	53
10. Iteration 3.....	54
10.1. Use Cases Covered.....	54
10.1.1. Extended Use Cases.....	54
10.1.2. System Sequence Diagrams.....	56
System Sequence Diagram 01	56
System Sequence Diagram 02:	57
10.1.3. Activity Diagram	58
11. Implementation details of Iteration 3	59
12. Iteration 4.....	61

13. Implementation details of Iteration 4.....	61
14. User Manual:.....	62
15. Technical Implementation.....	69
16. Testing.....	71
17. Technology Used.....	71
18. Poster.....	72
19. Conclusion.....	73
20. References.....	74

1. Introduction to the Project:

Social media and digital devices are a major part of living in this era. Where it has its perks it also comes with disadvantages. Most of the disadvantages arise from the security challenges associated. The Internet is not a secure place to work, one wrong move can wipe out all of your data. You are the most vulnerable when surfing the internet. Some of us have some idea of threats one has while some don't.

Attackers target weaker minds such as internet users, untrained employees, and illiterate people. The lesser someone knows about the threats the more weak and vulnerable they are. Hence easier to target and exploit. And while some of us know about some threats we don't know what to do after they've been attacked. How to counter the attack and most importantly how to mitigate the attack to ensure data security.

To cater to the above-mentioned problems we need a system a solution and a learning platform for the individuals who most often fall prey to such attacks. Secure Watch is a web application that aims to provide basic learning and practice platforms for individuals who want to protect themselves from attacks. Secure Watch provides basic cyber awareness to such individuals.

Our Project targets two basic audiences:

- Target audience 1: Untrained employees of organizations.
- Target audience 2: Internet users including illiterate people, children and aged people.

Each target audience is dealt with one of the four modules. We have covered four modules security terminologies, security threats, social engineering, and secure practices. The attacks are depicted by making exact clones of the platforms and the depiction of the attack can help the user virtually experience the attack and learn how to prevent it in the future.

Platforms targeted (clones): Facebook, Gmail, E-commerce website, blog website, Login portal, Banking Sites, and Donation sites.

We aim to educate these groups in a way that is fun and easy for them. Our platform will be providing lessons, graphical demonstrations, assessments, attack live simulations and mitigation techniques.

Sunday, 4th June, 2023.

2. Project vision:

2.1. Problem Statement

Organizations invest millions of dollars to keep their data safe but a single untrained employee can leave their organization vulnerable to an extensive list of attacks. In today's digital world, security of data is one of the most important step, leak of data can raise questions on the integrity and confidentiality of the organization in question. An organization can have numerous security measures installed in their systems none of which would help if the organization's employees are untrained. The untrained employees can intentionally or unintentionally engage in activities as simple as opening a mail, or logging into to a portal, by doing so they can risk their whole organization's integrity and reputation.

2.2. Business Opportunity

We would be providing free basic cyber lessons and lab simulations. However, we have a modular approach to our organization's data. In order to move on to the higher level of attack simulation labs, the users need to buy subscription and that is our main source of revenue.

2.3. Objectives

The main objectives of our applications are listed below.

- The main objective of application is to prevent cyber-attacks from happening.
- Providing a platform for lessons as well as assessments for hands on experience.
- Get daily updates on the current threat happening.
- Providing a platform to make user deal with any type of difficult situation.

2.4. Project Scope

SecureWatch revolutionizes the learning experience by bringing complex cybersecurity concepts to life through animated videos, assessments, and interactive practice labs. We aim to provide our users with a real-time experience of attacks, by our "learn by experience" model. We have a modular approach learn, experience, and mitigate. We provide a series of well-defined virtual labs to experience the attacks and find ways to tackle it. We aim to provide learners with real-world experience in tackling cyber-attacks.

Our project has been divided into the four phases such as:

September-October

- Module 1 content and assessment designing.
- Module 2 content and assessment designing.
- LMS design.
- Module 1 frontend

November-December

- Module 2 frontend.
- Module 3 and 4 lab content designing.
- Integration of Module 1 and 2 with LMS.

January-February

- Module 3 and 4 frontend.
- Integration of Module 3 and 4 with LMS.
- Module 1 and 2 backend.

March-April

- Integration of Module 3 and 4 with LMS.
- Module 1 and 2 backend.

2.5. Constraints

Here are the constraints associated with SecureWatch Web Application:

- The user will require an internet connection to access the material.
- Users with good scores can unlock the next levels.
- Users need to pay in order to access advanced labs.

2.6. Stakeholder Description

2.6.1. Stakeholder Summary:

The stakeholders of our application are:

- The user of our application:
 1. Cyber security untrained employees that can engage in activities to expose their organization's assets to threats.
 2. Internet users with zero knowledge about cyber security
 3. Aged people who fell victim easily to the social engineering techniques.
- The people associated with the service i.e. SecureWatch team.

2.6.2. Goals and Problems of Stakeholders:

1. Organization's reputation and data is at risk due to the untrained employees.
2. No training platforms for lay man in cybersecurity.
3. Internet users easily fall victim to the fraud calls and other social engineering techniques

3. Software Requirement Specifications:

3.1. List of Features

SecureWatch comprises of many high-level features that are listed below:

3.1.1. User Authentication/Sign up:

Our application allows user to login and save their activity in their account so that each time they login they don't have to start over and can continue from where they last left it.

3.1.2. User Interface:

The user interface consists of welcome screen that allows user to choose what they want to do, from watching a lesson and solving relevant activity to move to the next step of practicing their skills.

3.1.3. Video based Content:

SecureWatch provides video-based content for the users to learn about the basic to the advance concepts of cyber security.

3.1.4. Interface to practice your skills:

This interface allows the user to practice their learnings. This interface comprises of advance labs that give the user the experience to actually test what they have learned so far.

3.1.5. Manage profile:

The user can manage their profile. Update and add their details to their profile.

3.1.6. Interface to test their knowledge:

The users can attempt activities related to the lessons they have learned so far, this feature gives them an idea of their learning.

3.2. Functional Requirements:

The following are the functional requirements of our application:

3.2.1. Login/ Sign up:

The users should be able to register and login to the system

3.2.2. View Lessons:

The users should be able to watch the lessons designed.

3.2.3. View the number of lessons completed:

The users should be able to view the total number of the lessons they have completed so far.

3.2.4. Complete Activities:

The users should be able to complete the activities for the corresponding lessons.

3.2.5. View the number of activities completed:

The users should be able to view the total number of the activities they have completed so far.

3.2.6. Practice your knowledge:

The users should be able to practice their learnings on the labs designed.

3.2.7. View the labs progress:

The users can check how many labs they have completed so far.

3.3. Quality Attributes:

The following are the quality attributes of our application:

3.3.1. Maintainability:

Our application is easy to maintain. It is open to any new changes that you want to add.

3.3.2. Usability:

The application SecureWatch is user friendly. The interface is designed keeping in mind about the issues that the user might face and make interaction as easy as possible for the user.

3.3.3. Efficiency:

The application is efficient. The application efficiently utilizes the cpu memory and disk space during any point of the application

3.3.4. Availability:

The application is available for user at any point of the time. The features of the application can be accessed anytime. The user does not have to wait.

3.3.5. Security:

The user data in the application is completely safe. It is designed in a way to keep user data confidential and records safe.

3.4. Non-Functional Requirements:

Nonfunctional requirements are known as quality attributes; they describe common system characteristics of a system. The following are the non-functional requirements of our application:

3.4.1. Maintainability:

Our application is easy to maintain. It is open to any new changes that you want to add.

3.4.2. Usability:

The application SecureWatch is user friendly. The interface is designed keeping in mind about the issues that the user might face and make interaction as easy as possible for the user.

3.4.3. Efficiency:

The application is efficient. The application efficiently utilizes the CPU memory and disk space during any point of the application

3.4.4. Availability:

The application is available for user at any point of the time. The features of the application can be accessed anytime. The user does not have to wait.

3.4.5. Security:

The user data in the application is completely safe. It is designed in a way to keep user data confidential and records safe.

4. High-level Use Cases:

Use case 1:

UC01: USE CASE Create an Account	
Actors:	End-user
Type:	Primary
Description:	The user opens the application. Creates an account to use the features of the application. The account keeps a record of the activity of the end user.

Use Case UC01: Create an Account	
Scope:	The process of adding an activity to the application.
Level:	User goal level
Primary Actor:	Administrator
Stakeholders and interests:	<u>End-user:</u> Create an account to use the available features of the application.
Preconditions:	None
Success Guarantee:	Successful creation of user account.
Main success scenario:	1. The user provides the relevant required information necessary to create the account. 2. The system validates the provided information. 3. Stores the user data in the database. 4. The account is successfully created.
Extensions:	If the provided user data is not suitable an error message is displayed.
Special Requirements:	The user must have an internet connection when creating an account.
Technology and Data Variations list:	

Use case 2:

UC02: USE CASE Log into the System	
Actors:	End-user
Type:	Primary
Description:	The user opens the application. The user enters a username and password. The system verifies the user and lets the user enter the application.

Use Case UC02:	Log into the System
Scope:	The process of logging the user to the application.
Level:	User goal level
Primary Actor:	Administrator
Stakeholders and interests:	<u>End-user:</u> Successful login to the application.
Preconditions:	The user must have created an account.
Success Guarantee:	Successful login of the user account.
Main success scenario:	1. The user enters the credentials for logging in. 2. The system verifies the credentials. 3. The successful verification allows the user to log in.
Extensions:	1. If the provided user credentials are not verified an error message is displayed and the user is allowed to attempt login up to 3 times. 2. If the 3rd attempt is failed the user is given the option to reset the password or the username on the provided email.
Special Requirements:	The user needs internet connection to log in to the system.
Technology and Data Variations list:	

Use Case 3

UC03: USE CASE	Take Lesson
Actors:	End-user
Type:	Primary
Description:	The user enters the application after successful login. The user can take lessons on the basic and necessary terms of cyber security.

Use Case UC03:	Take level Lesson
Scope:	The user takes the lesson
Level:	User goal level
Primary Actor:	End-user
Stakeholders and interests:	<u>End-user:</u> The user takes the lesson and learns about them through the animated videos.
Preconditions:	The user must log in to their account
Success Guarantee:	The user completes the lesson they are taking
Main success scenario:	1. The user must log in to the account. 2. The user opens the lesson's features. 3. The user begins with the designated lesson. 4. The user finishes the lesson.
Extensions:	
Special Requirements:	The user should have internet connection.
Technology and Data Variations list:	

Use Case 4:

UC04: USE CASE Attempt Assessment	
Actors:	End-user
Type:	Primary
Description:	The user enters the application after successful login. The user attempts the designed corresponding assessment to the relevant lecture.

Use Case UC04: Attempt Assessment	
Scope:	The user completes the quiz
Level:	User goal level
Primary Actor:	End-user
Stakeholders and interests:	<u>End-user:</u> The user completes the quiz that helps them to test their knowledge about the relevant lesson.
Preconditions:	1. The user must log in to their account
	2. The user must have completed the lesson associated with the activity.
Success Guarantee:	The user completes the activity they are taking
Main success scenario:	1. The user after taking the lesson proceeds to the relevant assessment. 2. The user completes the assessment.
Extensions:	
Special Requirements:	The user should have internet connection.
Technology and Data Variations list:	

Use Case 5:

UC05: USE CASE	View Score
Actors:	End-user
Type:	Primary
Description:	The user enters the application after successful login. The user can see the score they have earned by completing the assessments.

Use Case UC05:	View Score
Scope:	The user watches lessons, solves assessment and view the score earned.
Level:	User goal level
Primary Actor:	End-user
Stakeholders and interests:	<u>End-user:</u> The user after completing assessments, can view how many points they have earned.
Preconditions:	1. The user must log in to their account
Success Guarantee:	The user views the score they earned.
Main success scenario:	The user looks at the scores.
Extensions:	
Special Requirements:	The user should have an internet connection.
Technology and Data Variations list:	

Use Case 06

UC06: USE CASE	Learn Mitigation Techniques
Actors:	End-user
Type:	Primary
Description:	The user enters the application after successful login. The user can see mitigation techniques corresponding to each lab at the end of it.

UC06: USE CASE	Learn Mitigation Techniques
Scope:	The user learns the mitigation techniques.
Level:	User goal level
Primary Actor:	End-user
Stakeholders and interests:	<u>End-user:</u> The user sees mitigation techniques of labs and gets to know how to take necessary steps to avoid cyber-attacks.
Preconditions:	1. The user must log in to their account
Success Guarantee:	The user learns the mitigation techniques.
Main success scenario:	The user learns the litigation techniques.
Extensions:	
Special Requirements:	The user should have an internet connection.
Technology and Data Variations list:	

Use Case 7:

UC07: USE CASE	Experience Practical Labs
Actors:	End-user
Type:	Primary
Description:	The user enters the application after successful login. The user can access the feature to solve labs.

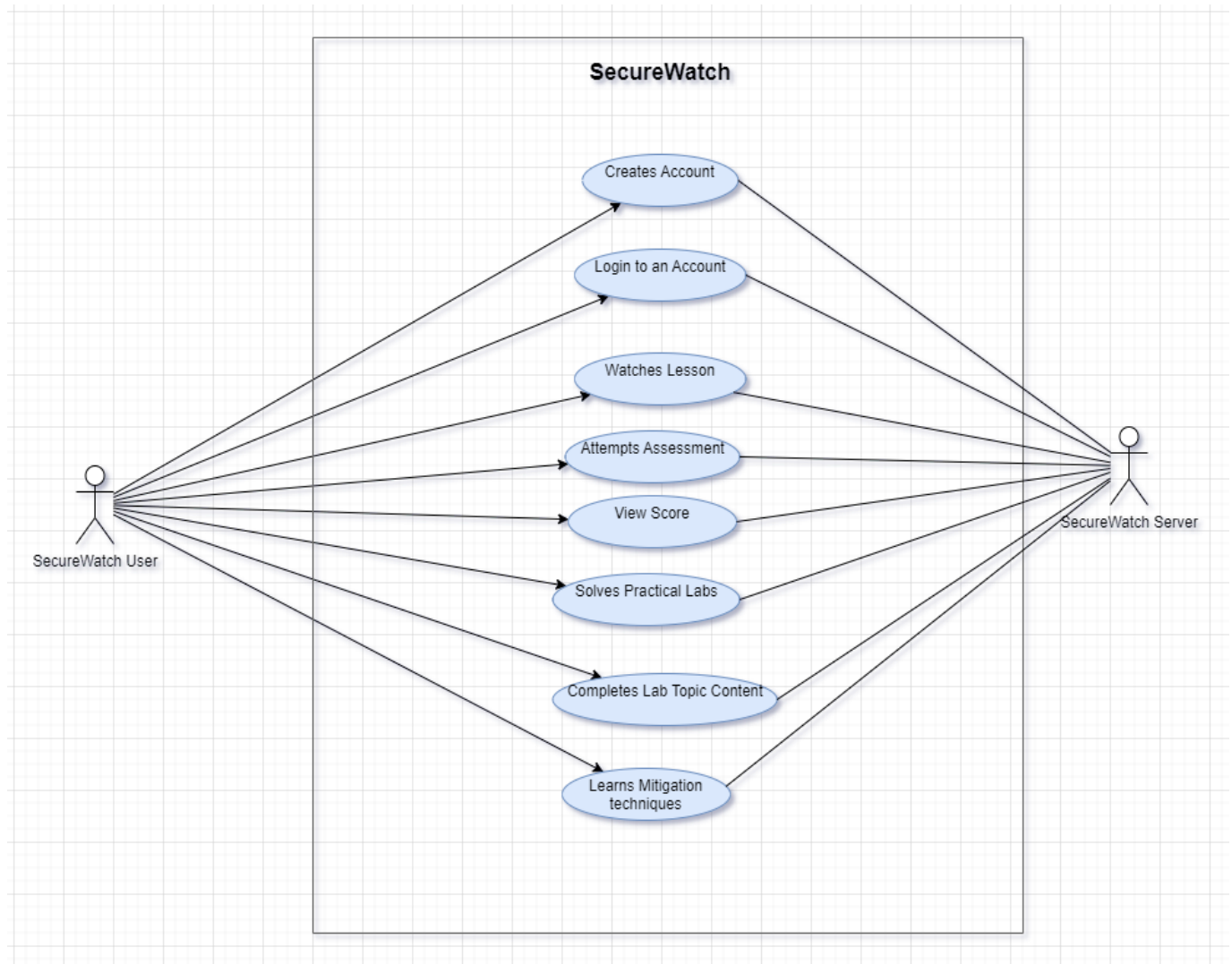
Use Case UC07:	Experience Practical Labs
Scope:	The user experiences labs
Level:	User goal level
Primary Actor:	End-user
Stakeholders and interests:	<u>End-user:</u> The user solves labs, the labs are designed for the people who have basic knowledge about the cyber world provide them with an experience that how cyber-attacks are present everywhere.
Preconditions:	1. The user must log in to their account
	2. The user must have basic knowledge about the cyber world.
Success Guarantee:	The user completes solving the lab
Main success scenario:	1. The user logs in to the application and enters the category of labs. 2. The user can access the labs designed. 3. The user experiences the labs. 4. The user takes help from the hints option available when solving the lab.
Extensions:	
Special Requirements:	The user should have an internet connection.
Technology and Data Variations list:	

Use Case 08:

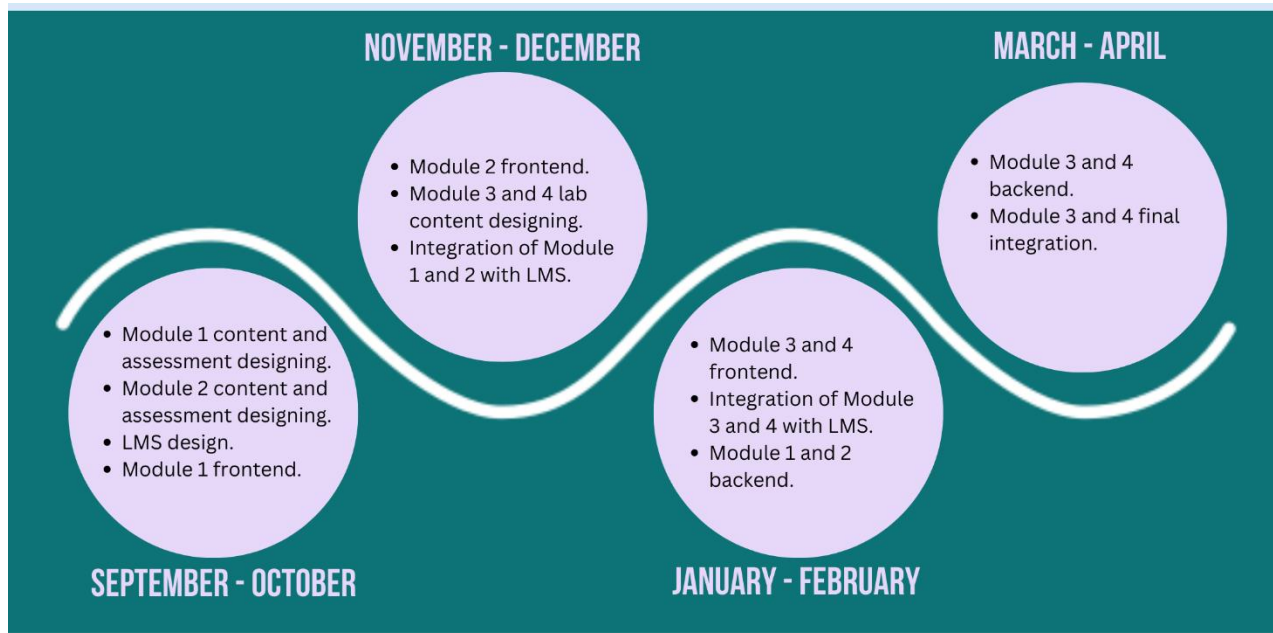
UC08: USE CASE	Complete lab Topic Content
Actors:	End-user
Type:	Primary
Description:	The user enters the application after successful login. The user access the feature to solve labs and after following the steps the user reaches the end of the lab and completes the lab.

UC08: USE CASE	Complete lab Topic Content
Scope:	The user completes the lab topic.
Level:	User goal level
Primary Actor:	End-user
Stakeholders and interests:	<u>End-user:</u> The user tries to solve labs and after following the steps the user reaches the end of the lab and completes the lab..
Preconditions:	1. The user must log in to their account
Success Guarantee:	The user completes the lab.
Main success scenario:	The user completes the lab topic.
Extensions:	
Special Requirements:	The user should have an internet connection.
Technology and Data Variations list:	

4.1. Use Case Diagram



5. Iteration Plan



6. Iteration 1

6.1. Use cases covered

6.1.1. Extended Use Cases

Extended Use Case 01

UC01: USE CASE Create an Account	
Actors:	End-user
Type:	Primary
Description:	The user opens the application. Creates an account to use the features of the application. The account keeps a record of the activity of the end user.

Use Case UC01: Create an Account	
Scope:	The process of adding an activity to the application.
Level:	User goal level
Primary Actor:	Administrator
Stakeholders and interests:	<u>End-user:</u> Create an account to use the available features of the application.
Preconditions:	None
Success Guarantee:	Successful creation of user account.
Main success scenario:	1. The user provides the relevant required information necessary to create the account. 2. The system validates the provided information. 3. Stores the user data in the database. 4. The account is successfully created.
Extensions:	If the provided user data is not suitable an error message is displayed.
Special Requirements:	The user must have an internet connection when creating an account.
Technology and Data Variation	

Extended Use Case 02

UC02: USE CASE Log into the System	
Actors:	End-user
Type:	Primary
Description:	The user opens the application. The user enters a username and password. The system verifies the user and lets the user enter the application.

Use Case UC02: Log into the System	
Scope:	The process of logging the user to the application.
Level:	User goal level
Primary Actor:	Administrator
Stakeholders and interests:	<u>End-user:</u> Successful login to the application.
Preconditions:	The user must have created an account.
Success Guarantee:	Successful login of the user account.
Main success scenario:	4. The user enters the credentials for logging in. 5. The system verifies the credentials. 6. The successful verification allows the user to log in.
Extensions:	3. If the provided user credentials are not verified an error message is displayed and the user is allowed to attempt login up to 3 times. 4. If the 3rd attempt is failed the user is given the option to reset the password or the username on the provided email.
Special Requirements:	The user needs internet connection to log in to the system.
Technology and Data Variations list:	

Extended Use Case 03

UC03: USE CASE	Take Lesson
Actors:	End-user
Type:	Primary
Description:	The user enters the application after successful login. The user can take lessons on the basic and necessary terms of cyber security.

Use Case UC03:	Take level Lesson
Scope:	The user takes the lesson
Level:	User goal level
Primary Actor:	End-user
Stakeholders and interests:	<u>End-user:</u> The user takes the lesson and learns about them through the animated videos.
Preconditions:	The user must log in to their account
Success Guarantee:	The user completes the lesson they are taking
Main success scenario:	1. The user must log in to the account. 2. The user opens the lesson's features. 3. The user begins with the designated lesson. 4. The user finishes the lesson.
Extensions:	
Special Requirements:	The user should have internet connection.
Technology and Data Variations list:	

6.1.2. System Sequence Diagrams

System Sequence Diagram 01

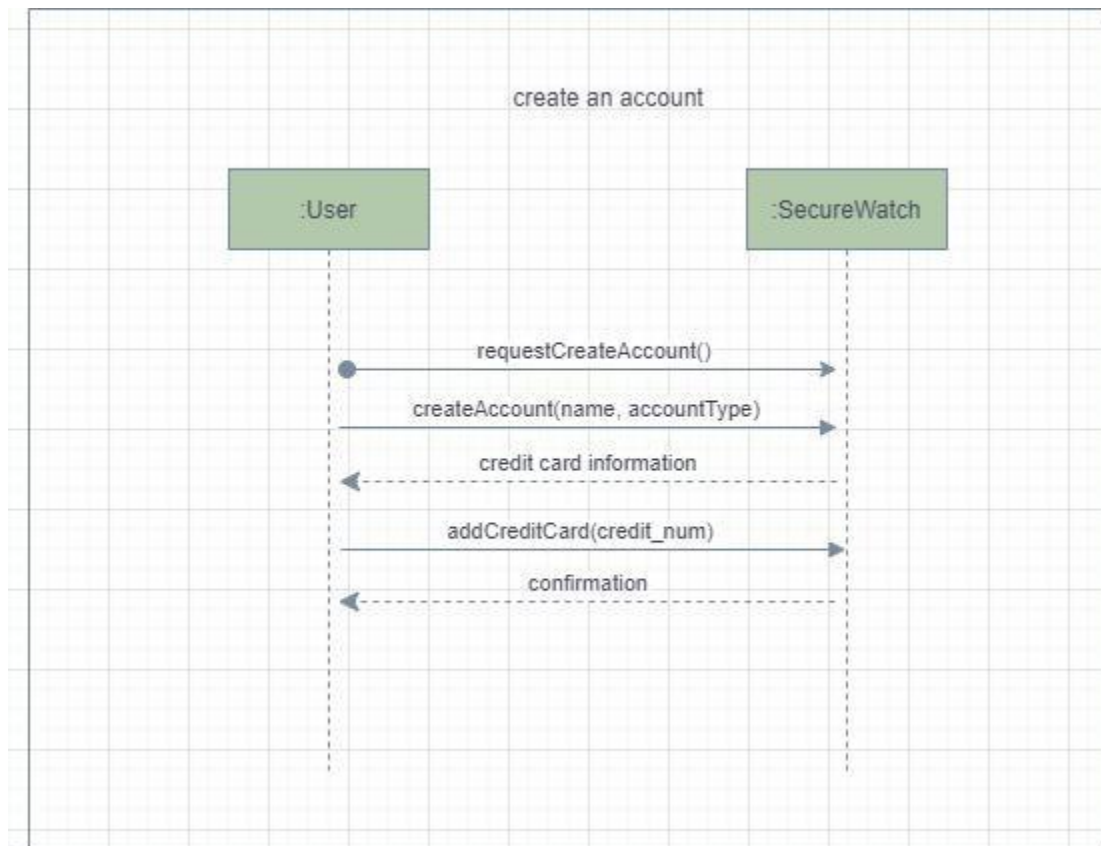
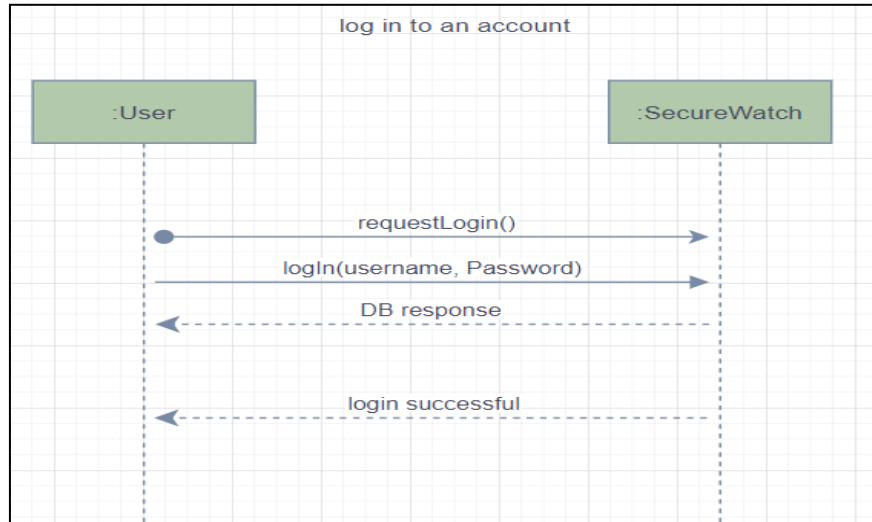
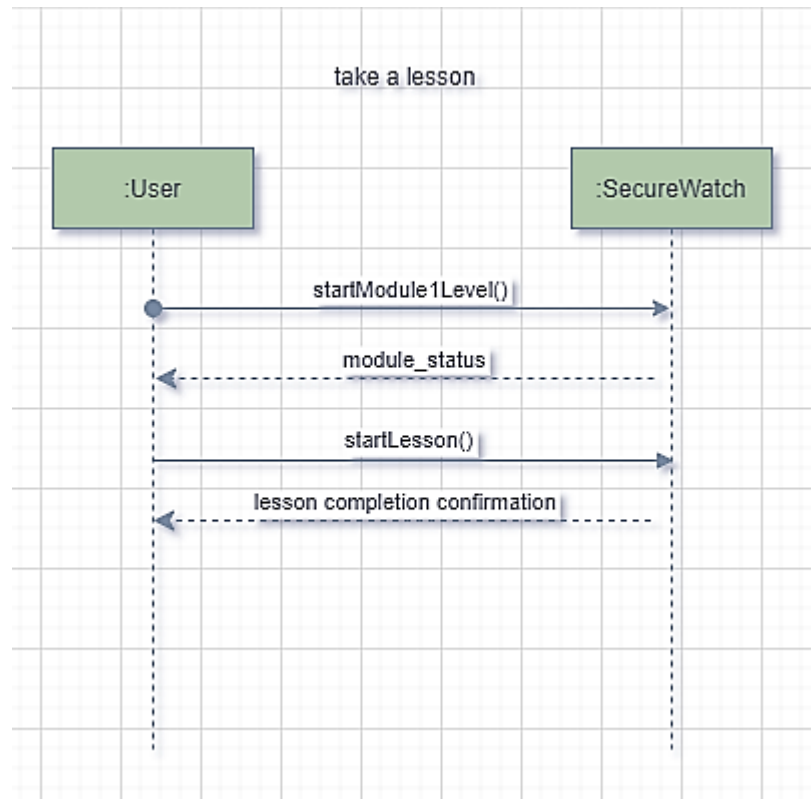


FIGURE 01 SSD CREATE ACCOUNT

System Sequence Diagram 02**FIGURE 02 SSD LOG-IN**

System Sequence Diagram 03**FIGURE 03 SSD: TAKE LESSON**

6.1.3. Activity Diagrams

Activity Diagram 01

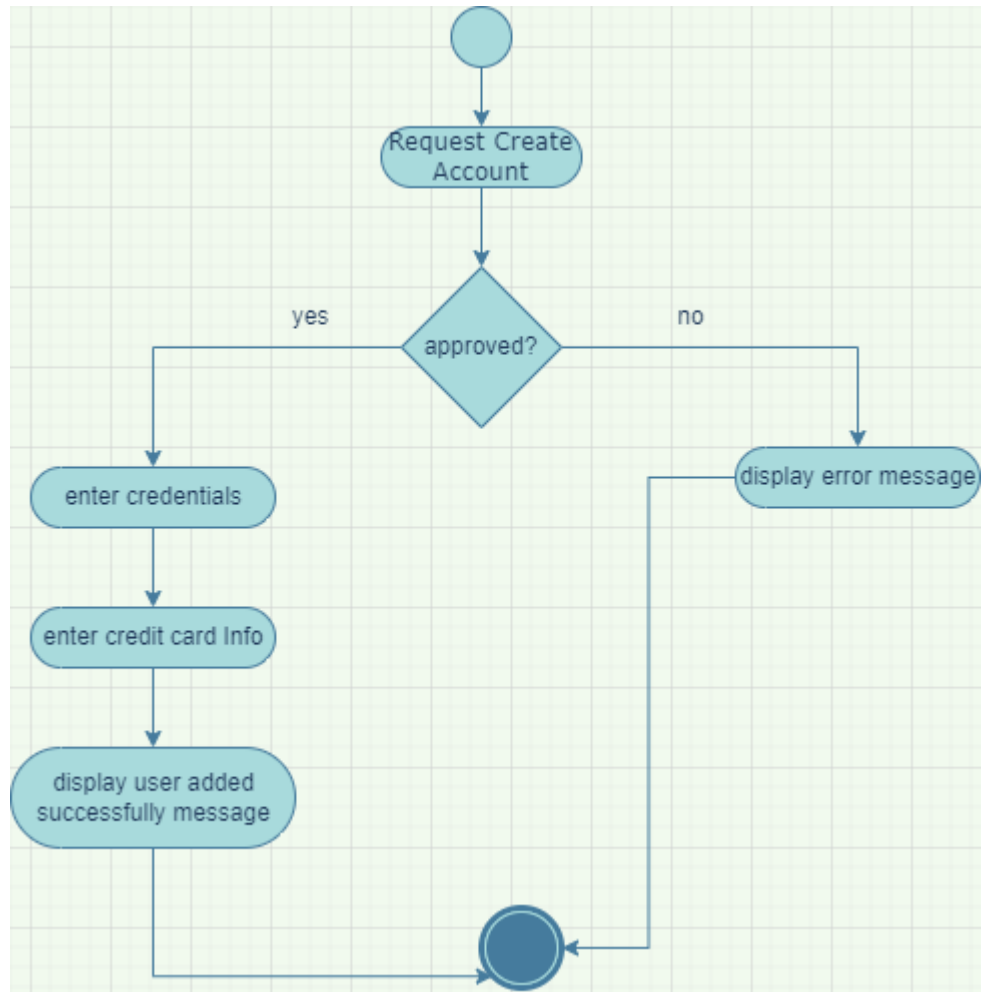
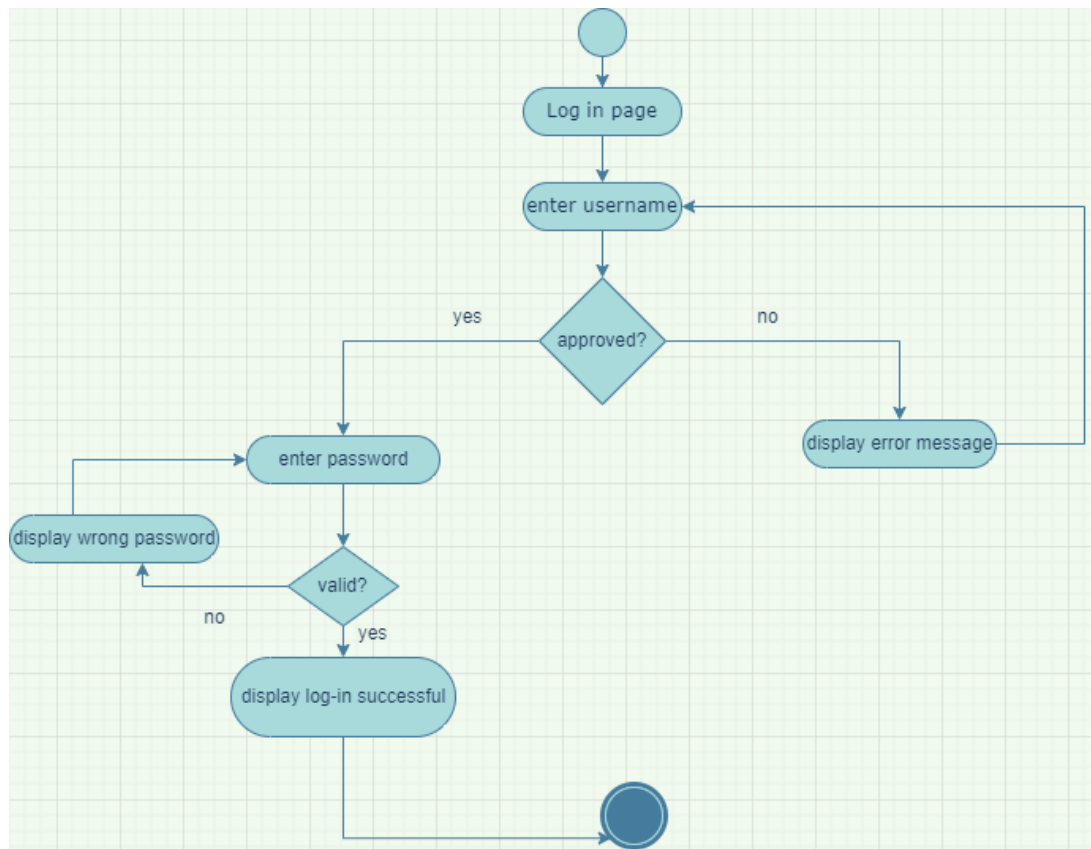
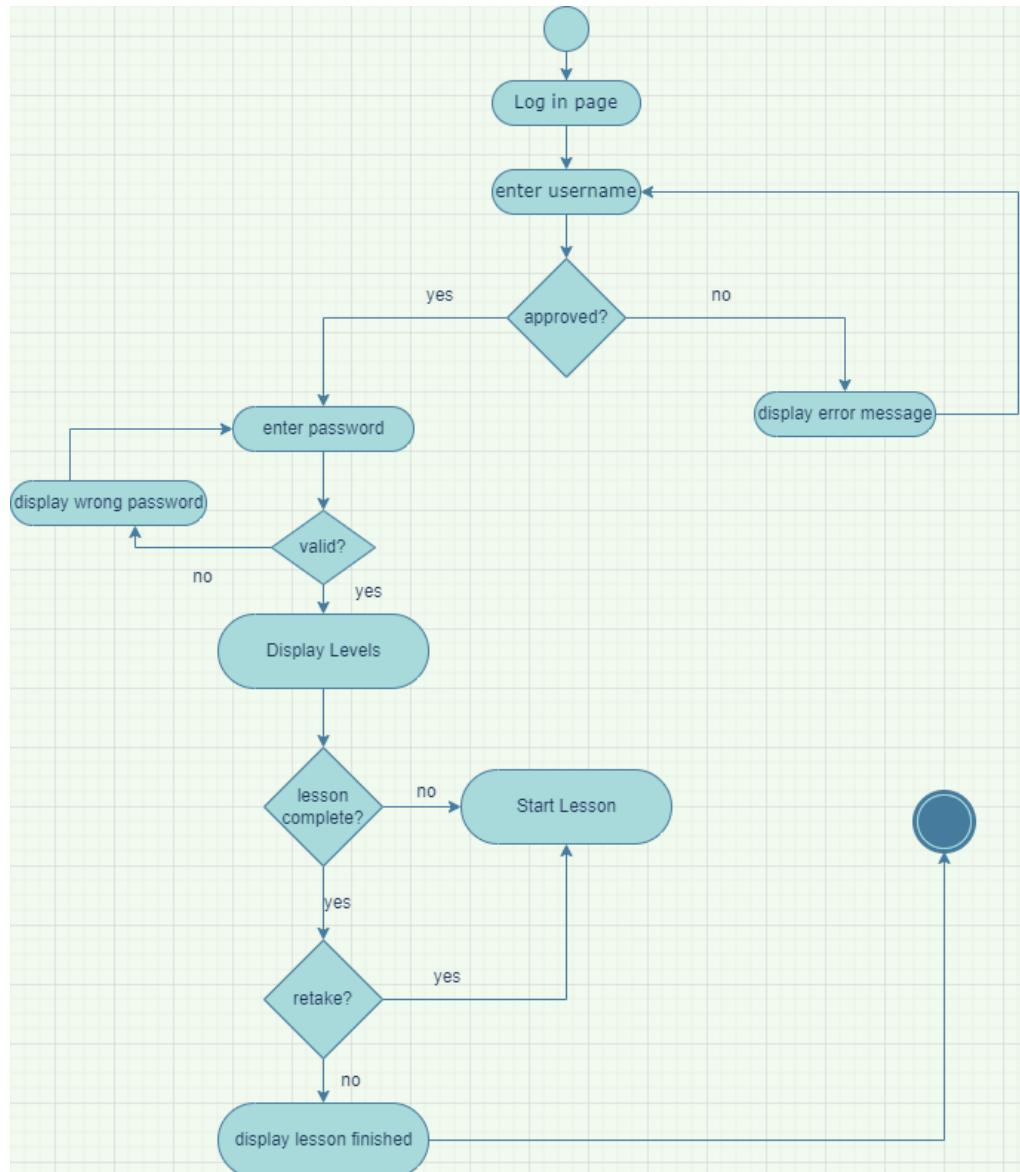
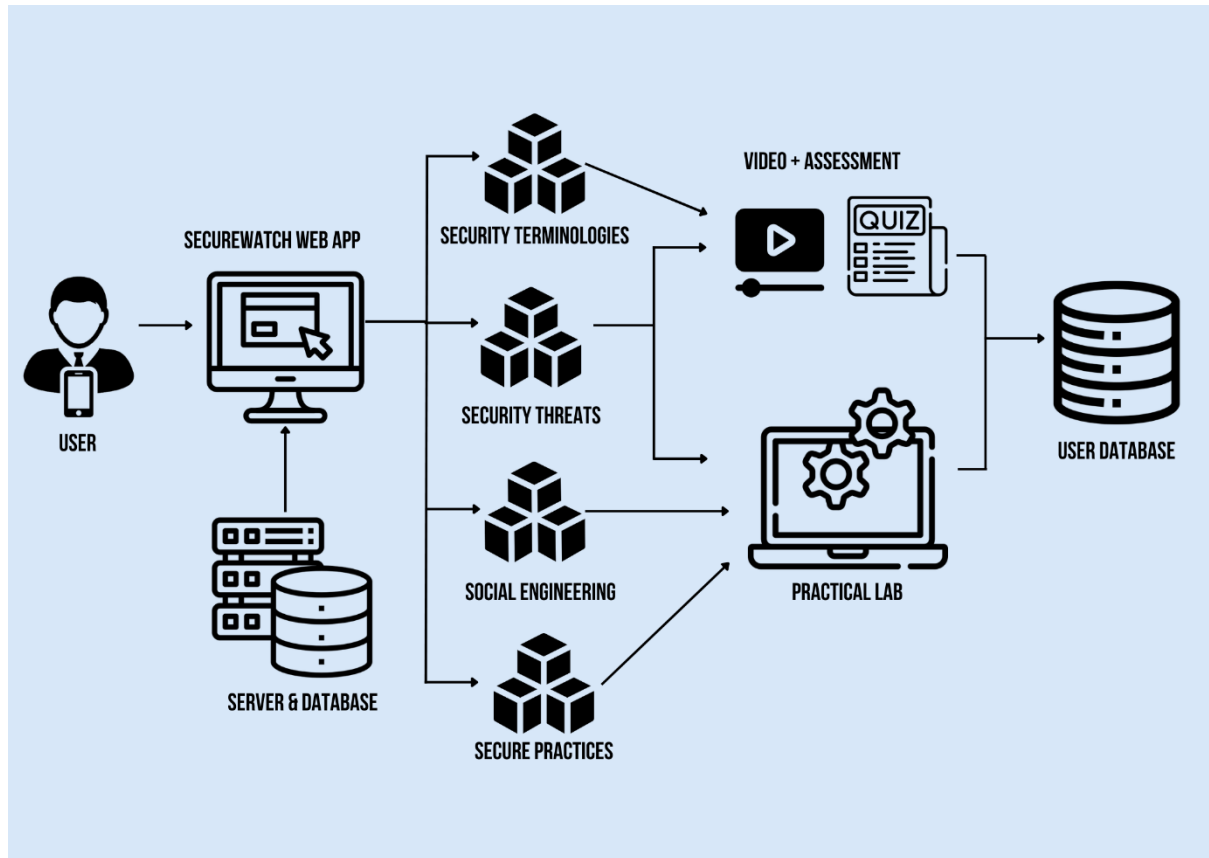


FIGURE 01 CREATE ACCOUNT

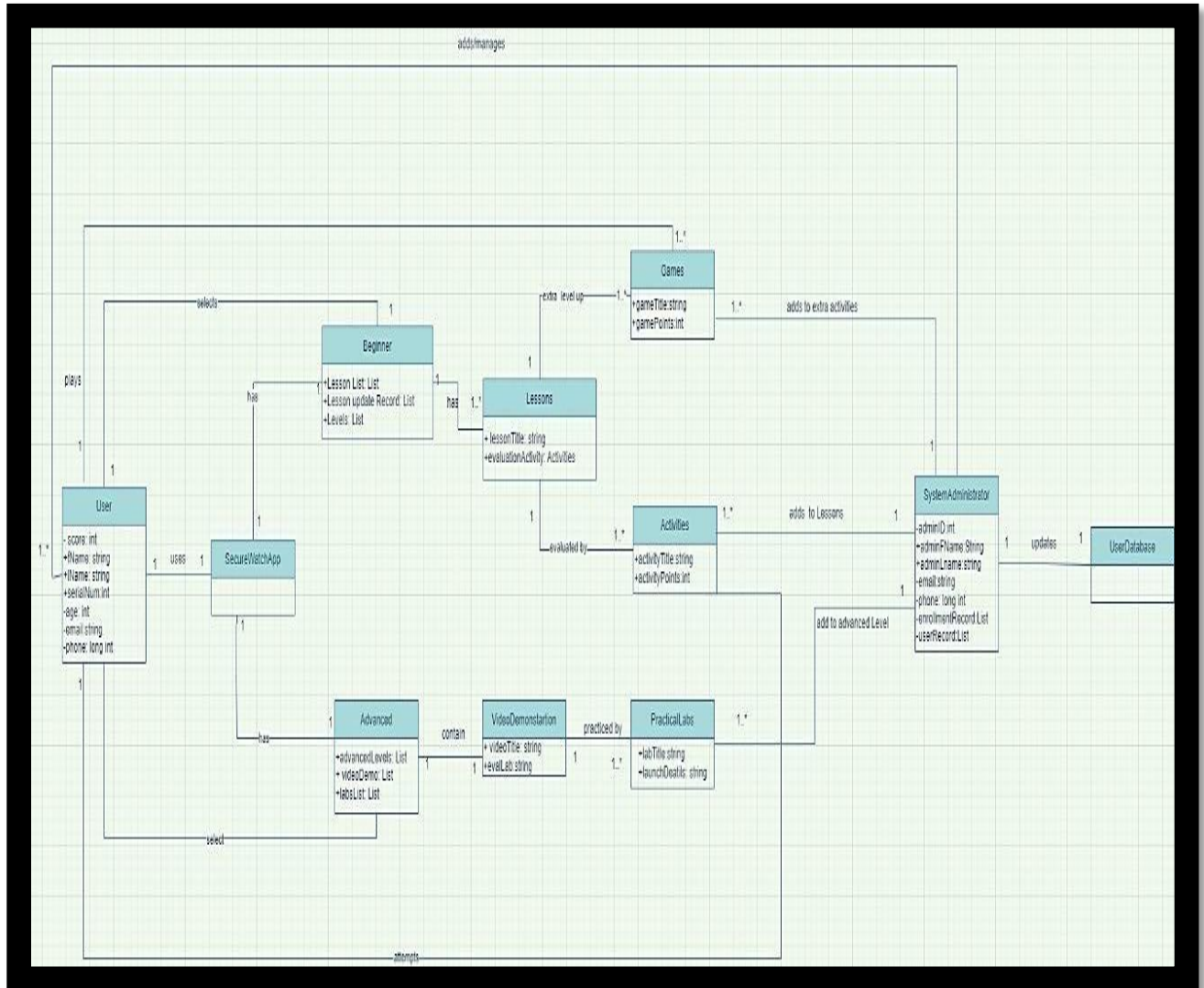
**Activity Diagram 02****FIGURE 02 LOG IN**

Activity Diagram 03**FIGURE 03 TAKE LESSON**

6.2. Architecture Diagram



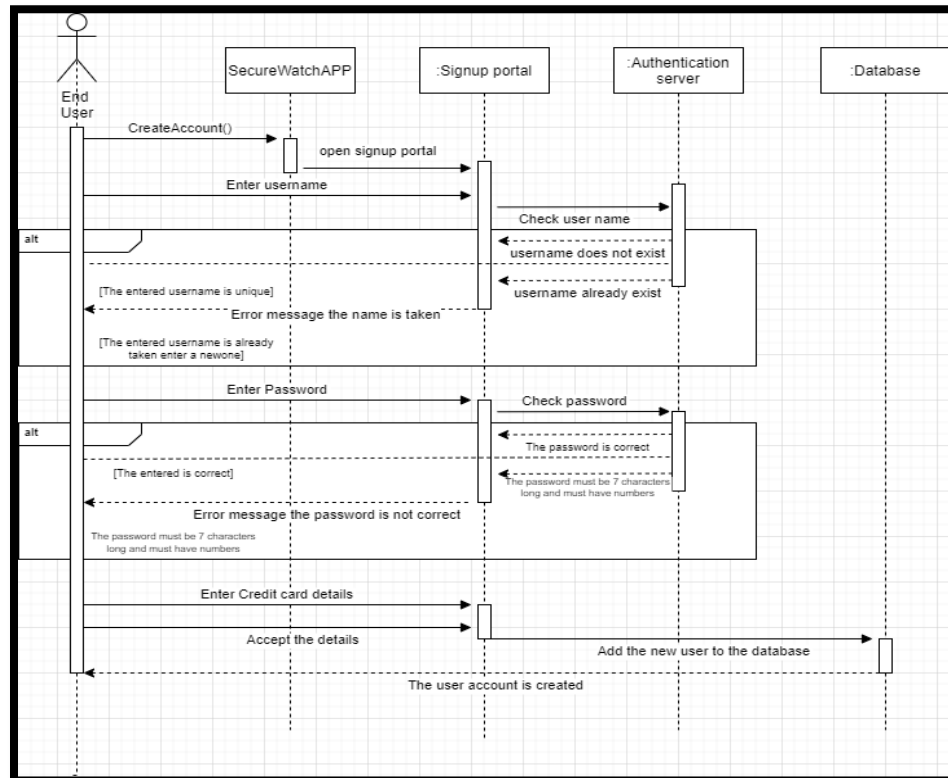
6.3. Domain Model Diagram



6.4. Sequence Diagrams

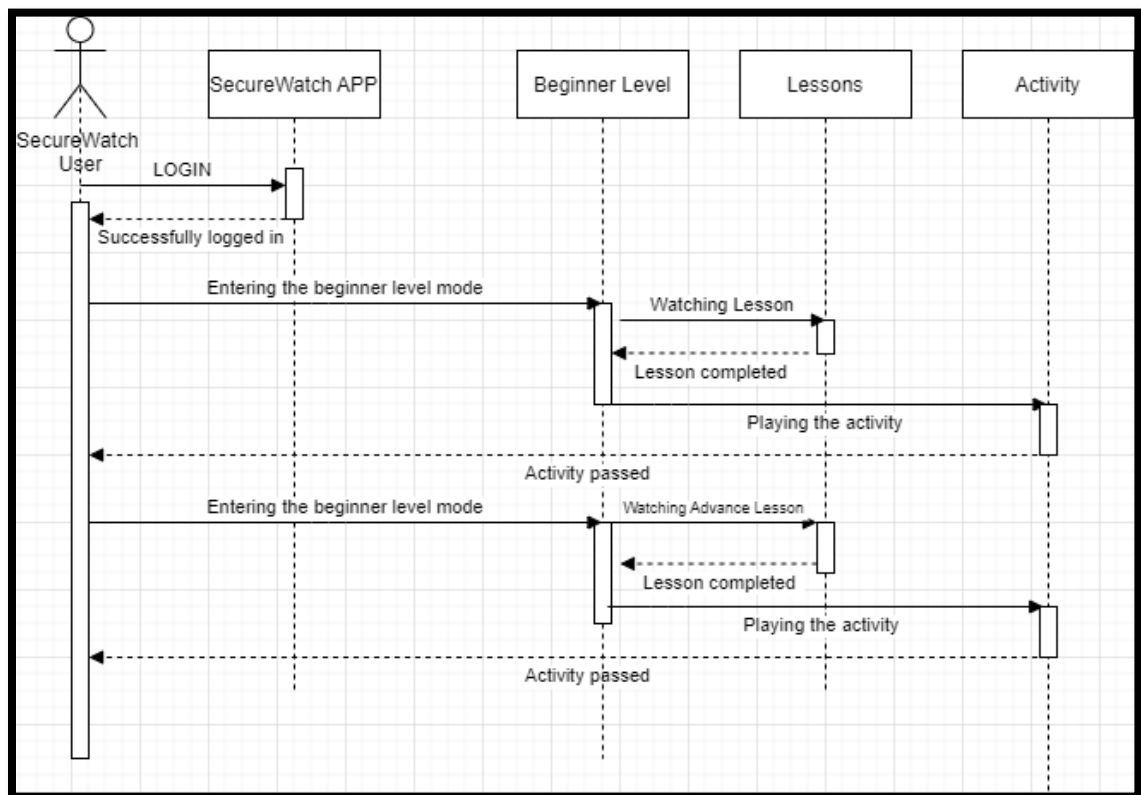
6.4.1. Scenario 1:

The user enters the application and wants to create account. The user enters user name and password, the app verifies the username and password. Then the user enters credit card details and the account is created.



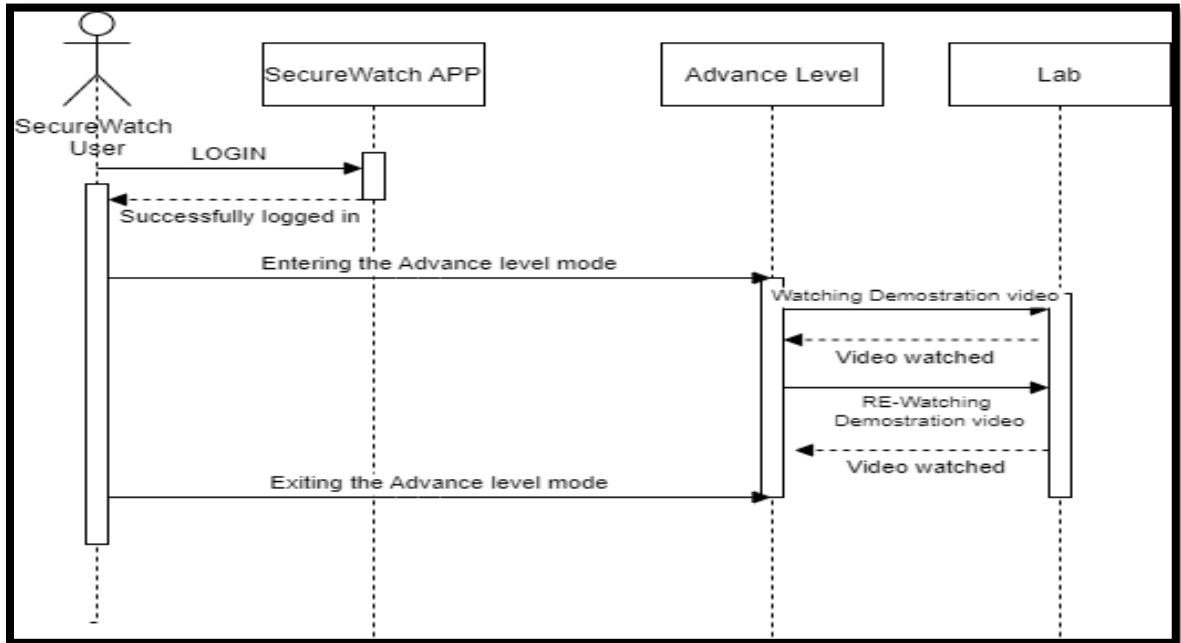
6.4.2. Scenario 2:

The user logs in and watches lesson, completes the corresponding activity and proceeds to the next ones.



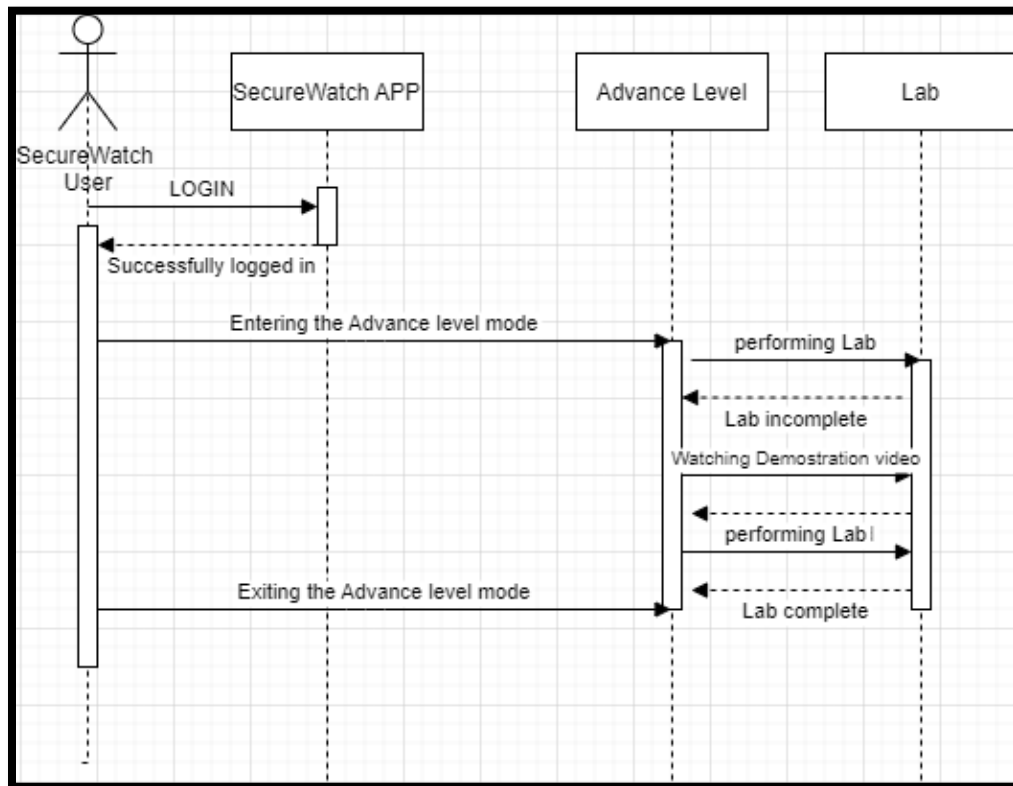
6.4.3. Scenario 3:

The user logs in and watches lab demonstration video twice.



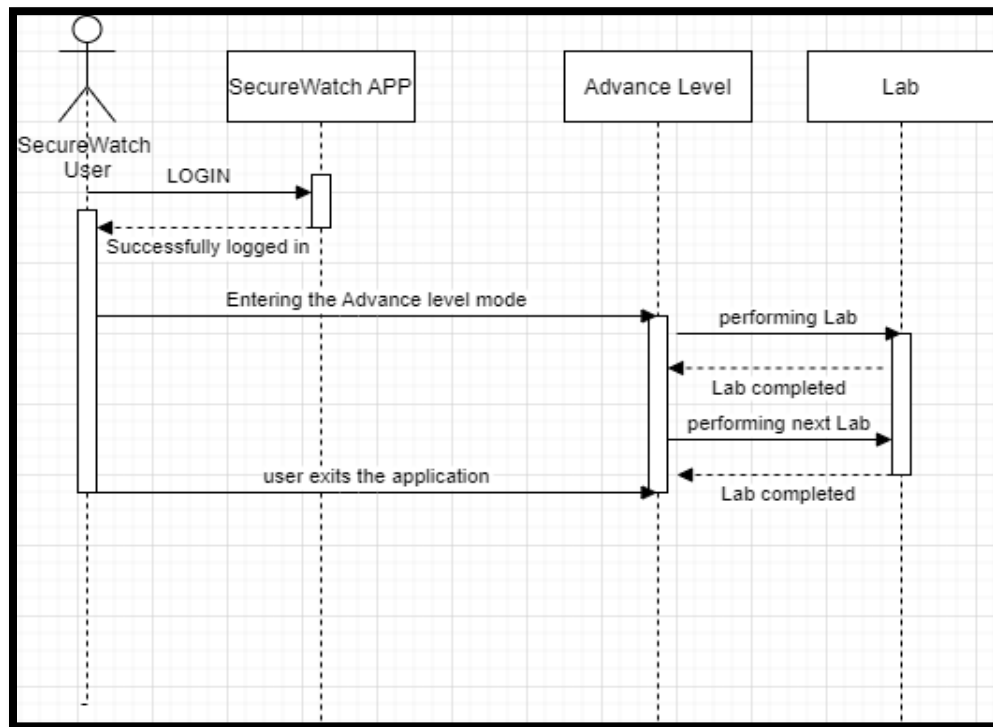
6.4.4. Scenario 4:

The user logs in, tries to complete the lab but was not able to. So the user watches the video and tries again. The user then completes the lab.



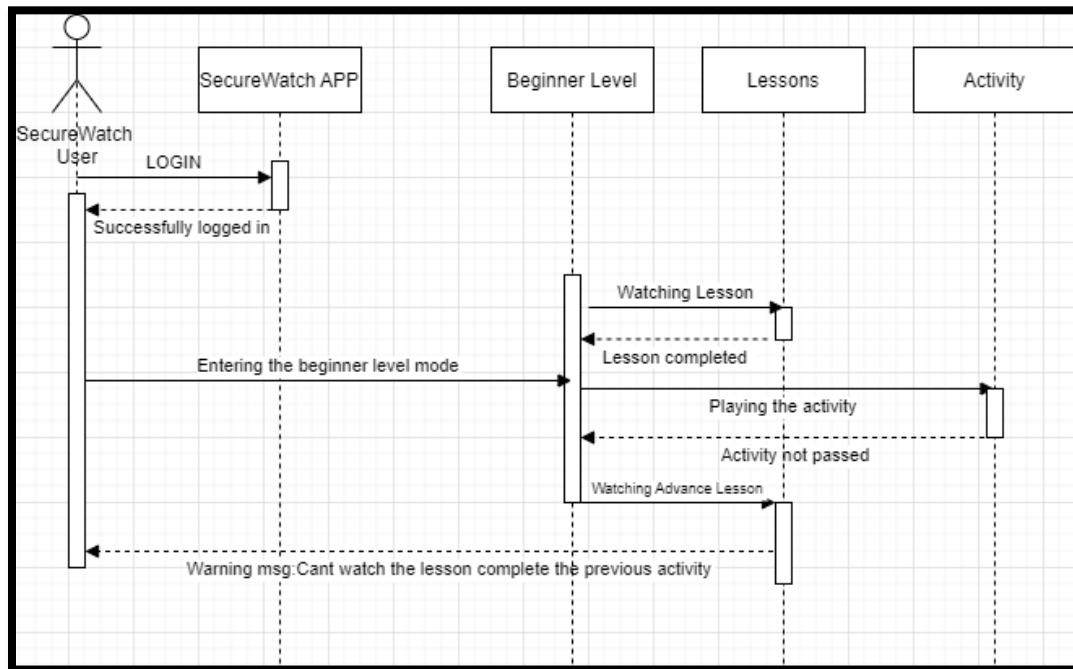
6.4.5. Scenario 5:

The user completed lab and proceeds to the next ones.



6.4.6. **Scenario 6:**

The user enters the beginner mode. The user watches the lesson and tries to complete the activity. But was not able to and watches the lesson again.



7. Implementation Details for Iteration 01

In this iteration the points to be completed were module 1 content designing. Module 1 is focused on security terminologies that means all the basic security terminology information that is pre-conditions of getting into to the cybersecurity training. We have designed this module in two parts first one being video lesson that video that covered the concepts used in the field of cybersecurity to describe and address various threats, risks, and protective measures. The designing part started with engaging introduction that highlights the topic, followed by real-world examples of scenarios which could lead to vulnerabilities to the system. The content was designed using Animaker, characters in the video represented real-world victims and attackers in the cases of different attack scenarios. The video content is designed that by the end of it the users are urged to adopt secure practices and learn about them.

We also designed Learning Management system frontend that is the basic front-end that included navigation bar, side bar, introductory description, designing and routes to modules.

Front-end to Module 1 was also completed it included a page to hold the video and navigation to the assessment. The assessment covered all the topics in the video and it has Multiple Choice Questions. After completion the user is presented with score in the assessment.

8. Iteration 2

8.1. Use cases covered

8.1.1. Extended Use Cases

Extended Use Case 01

UC04: USE CASE	Attempt Assessment
Actors:	End-user
Type:	Primary
Description:	The user enters the application after successful login. The user attempts the designed corresponding assessment to the relevant lecture.

Use Case UC04:	Attempt Assessment
Scope:	The user completes the quiz
Level:	User goal level
Primary Actor:	End-user
Stakeholders and interests:	<u>End-user:</u> The user completes the quiz that helps them to test their knowledge about the relevant lesson.
Preconditions:	1. The user must log in to their account
	2. The user must have completed the lesson associated with the activity.
Success Guarantee:	The user completes the activity they are taking
Main success scenario:	1. The user after taking the lesson proceeds to the relevant assessment. 2. The user completes the assessment.
Extensions:	
Special Requirements:	The user should have internet connection.
Technology and Data Variations list:	

Extended Use Case 02

UC05: USE CASE View Score	
Actors:	End-user
Type:	Primary
Description:	The user enters the application after successful login. The user can see the score they have earned by completing the assessments.

Use Case UC05: View Score	
Scope:	The user watches lessons, solves assessment and view the score earned.
Level:	User goal level
Primary Actor:	End-user
Stakeholders and interests:	<u>End-user:</u> The user after completing assessments, can view how many points they have earned.
Preconditions:	1. The user must log in to their account
Success Guarantee:	The user views the score they earned.
Main success scenario:	The user looks at the scores.
Extensions:	
Special Requirements:	The user should have an internet connection.
Technology and Data Variations list:	

Extended use case 03:

UC07: USE CASE Experience Practical Labs	
Actors:	End-user
Type:	Primary
Description:	The user enters the application after successful login. The user can access the feature to solve labs.

Use Case UC07: Experience Practical Labs	
Scope:	The user experiences labs
Level:	User goal level
Primary Actor:	End-user
Stakeholders and interests:	<u>End-user:</u> The user solves labs, the labs are designed for the people who have basic knowledge about the cyber world provide them with an experience that how cyber-attacks are present everywhere.
Preconditions:	1. The user must log in to their account 2. The user must have basic knowledge about the cyber world.
Success Guarantee:	The user completes solving the lab
Main success scenario:	1. The user logs in to the application and enters the category of labs. 2. The user can access the labs designed. 3. The user experiences the labs. 4. The user takes help from the hints option available when solving the lab.
Extensions:	
Special Requirements:	The user should have an internet connection.
Technology and Data Variations list:	

8.1.2. System Sequence Diagrams

System Sequence Diagram 01

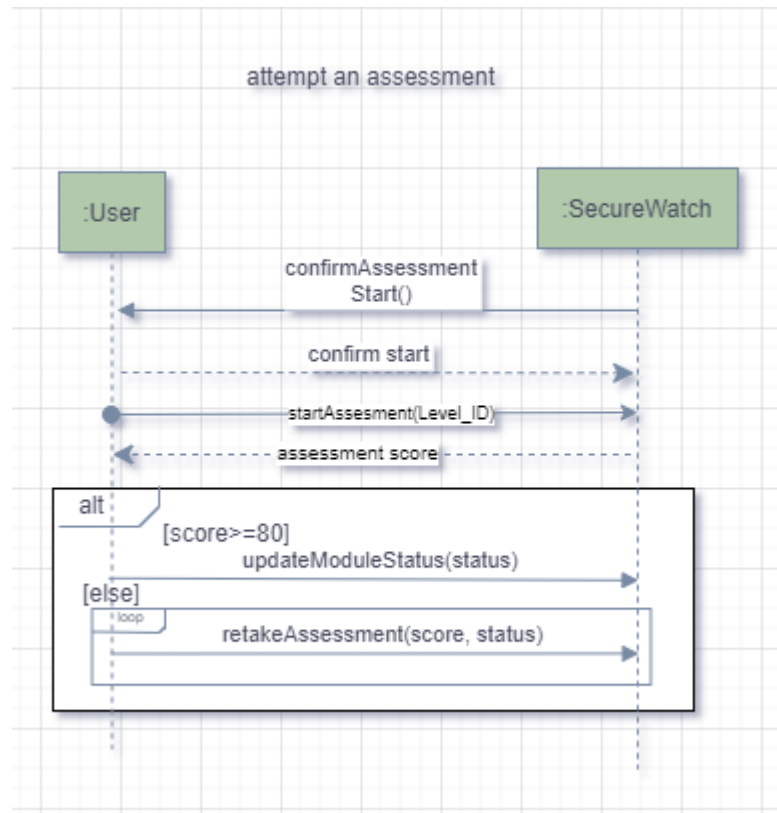
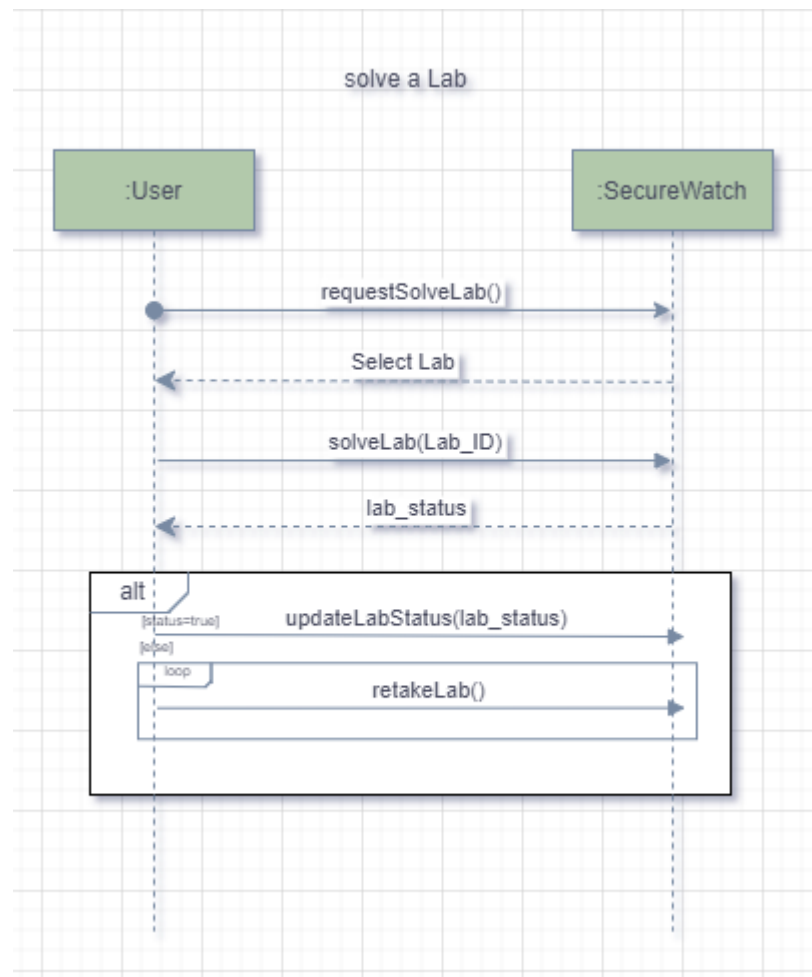


FIGURE 01 ATTEMPT ASSESSMENT

System Sequence Diagram 02**FIGURE 01 SSD SOLVE LAB**

8.1.3. Activity Diagrams

Activity Diagram 01

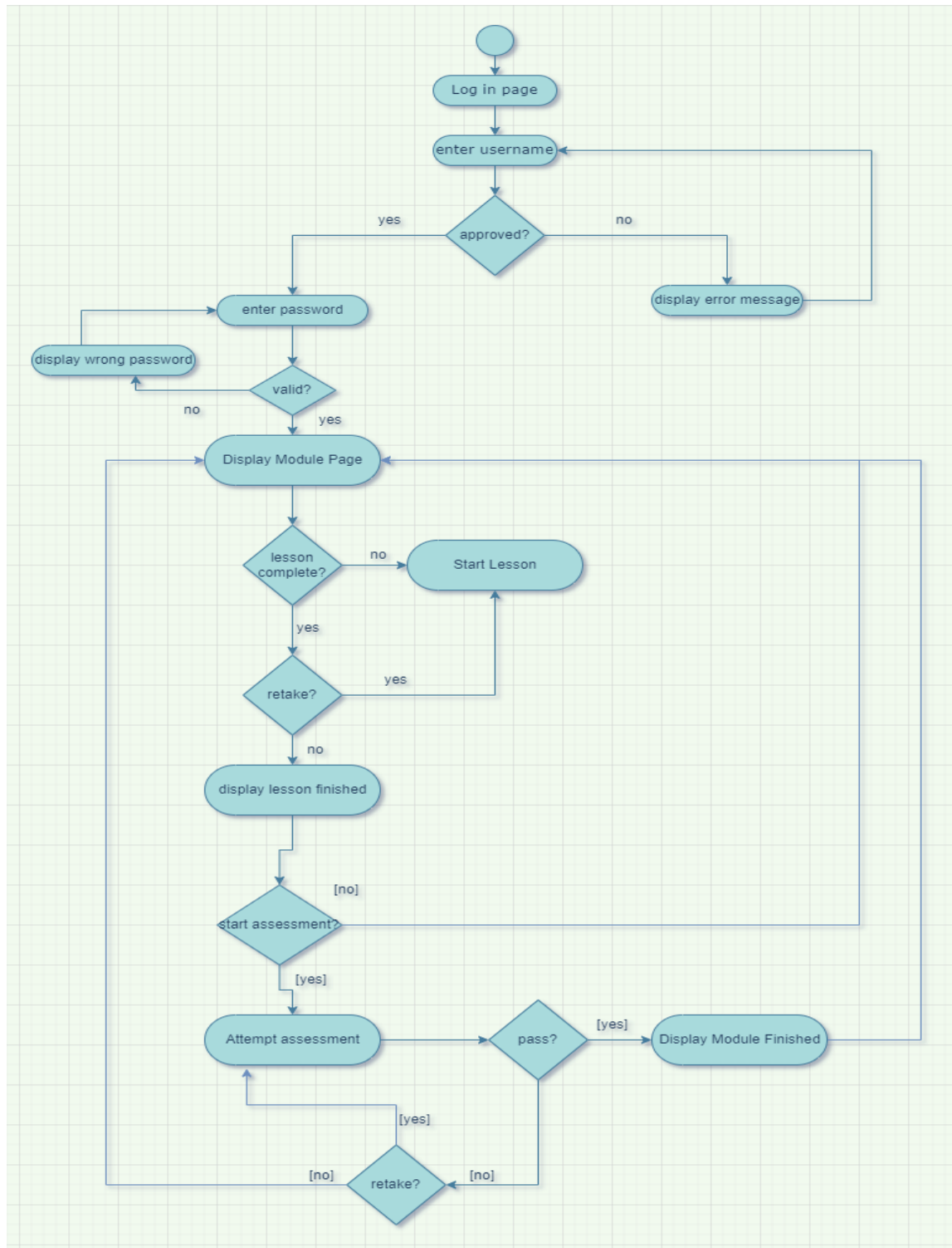
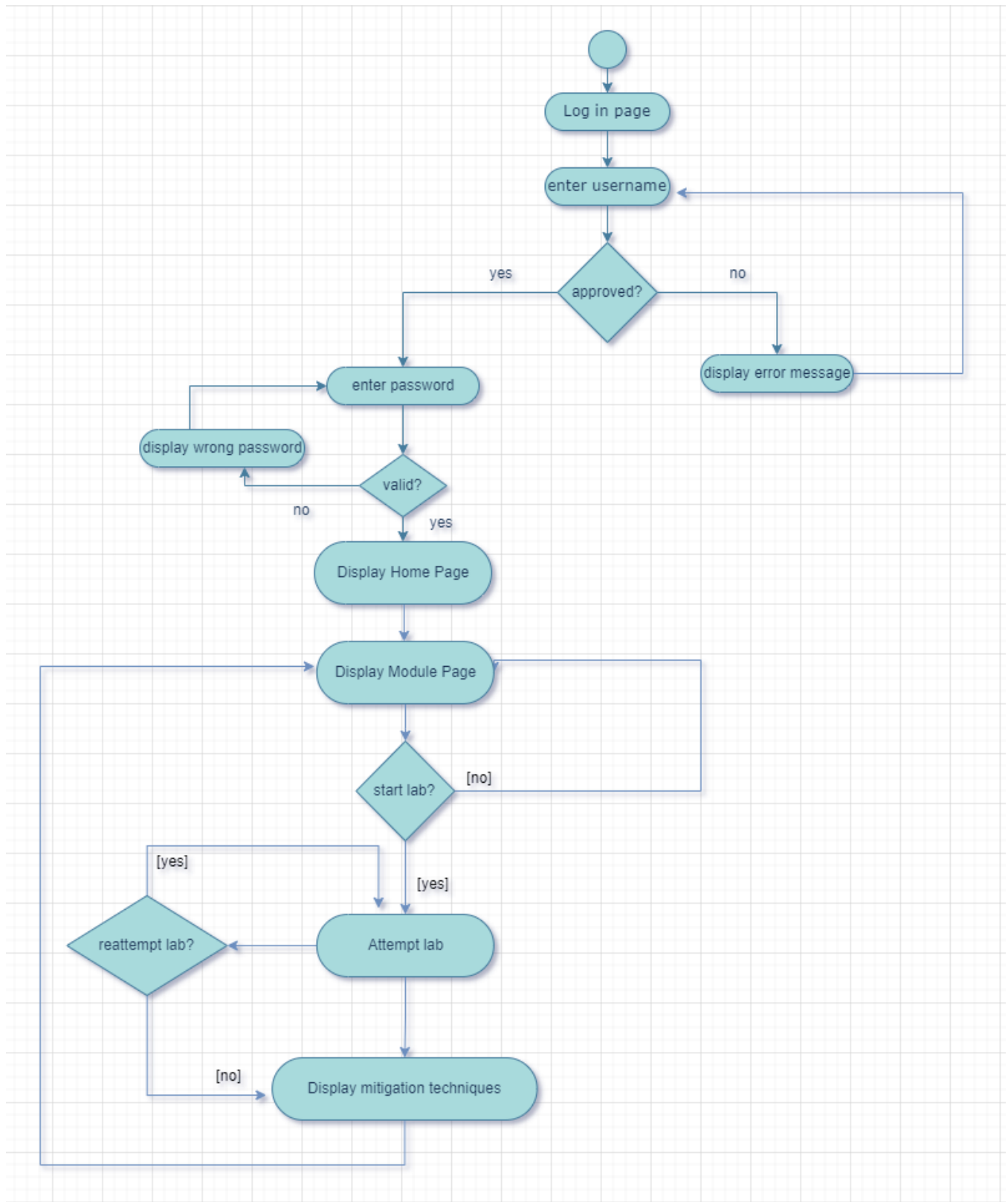
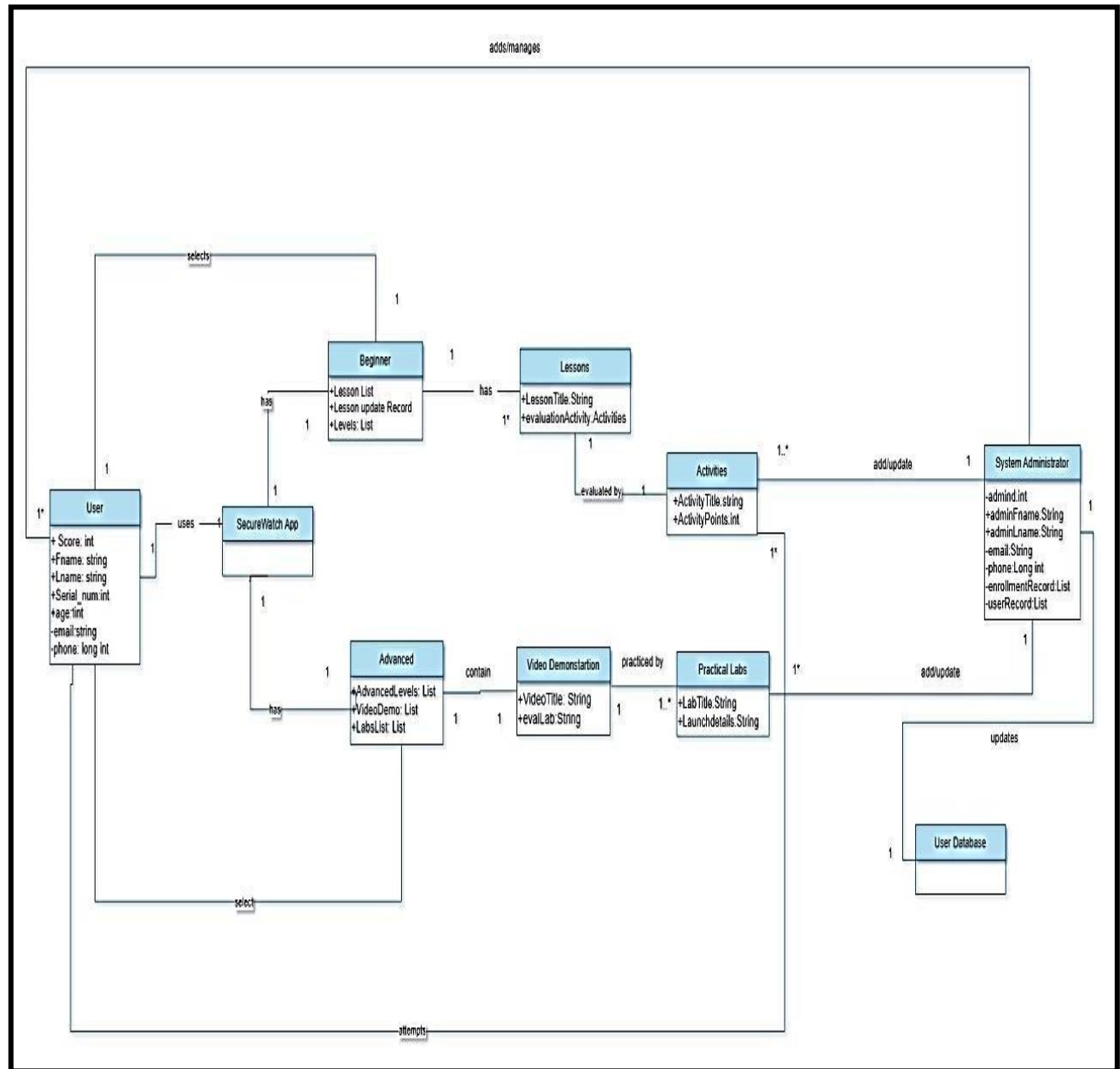


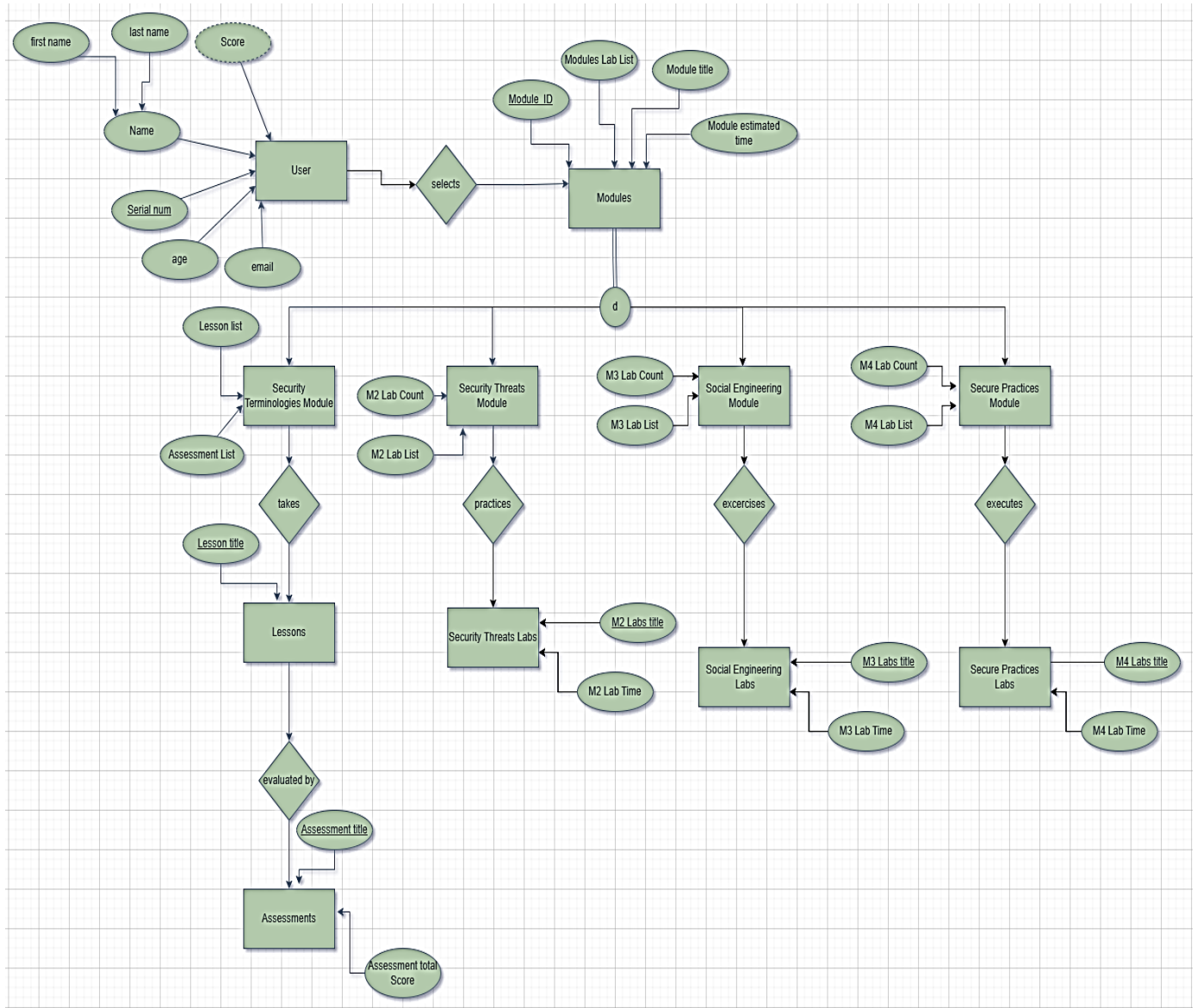
FIGURE 1 MODULE 1 ACTIVITY

Activity Diagram 02**FIGURE 2 ATTEMPT LAB**

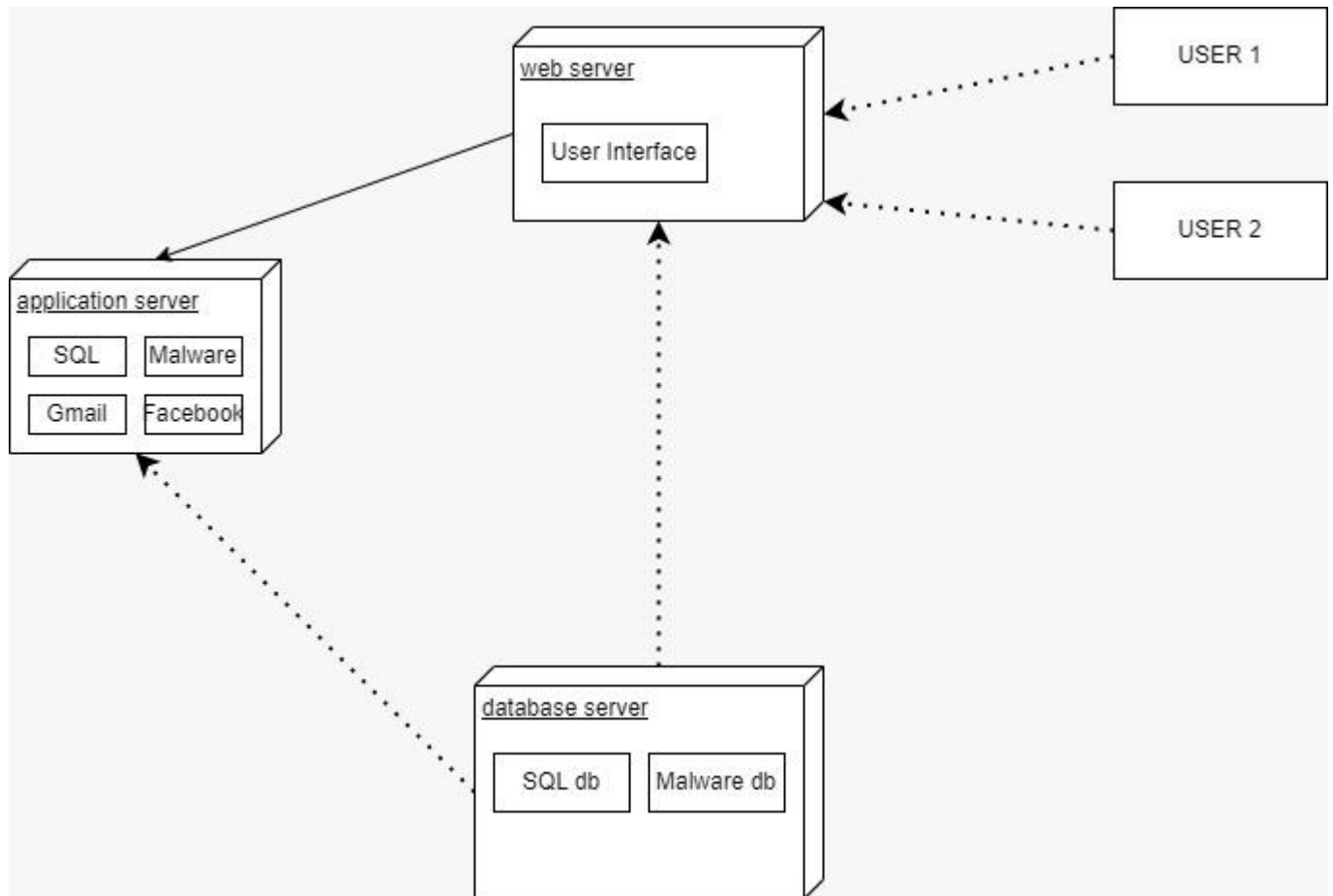
8.1.4. Domain Model Diagram



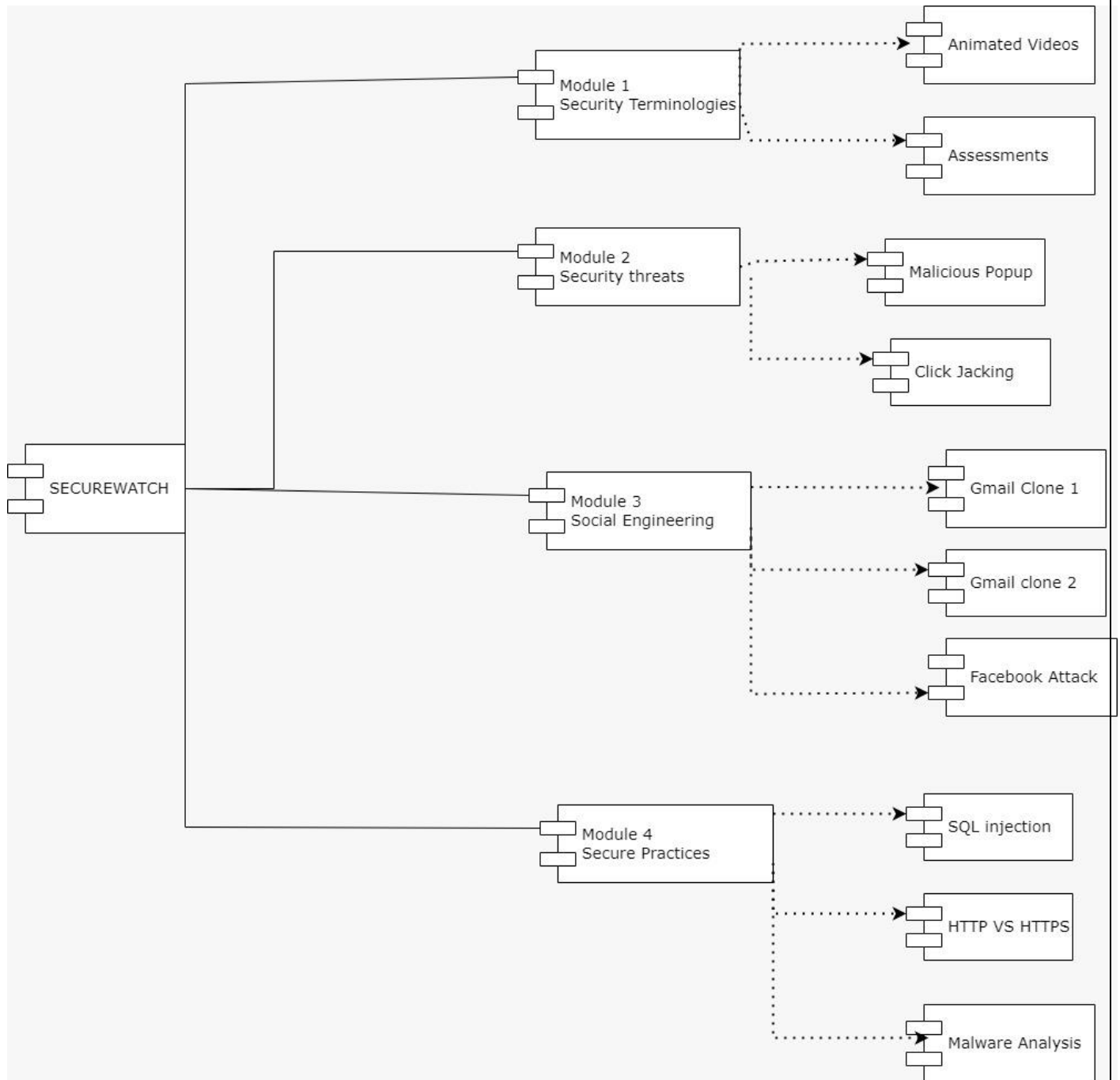
8.2. Entity Relationship Diagram



8.3. Deployment Diagram



8.4. Component Diagram



9. Implementation Details for Iteration 02

In this iteration, we were supposed to implement lab simulations each implanted with a different backend strategy. Here I will be explaining the backend and front strategy of the labs implemented.

We have covered the implementation of Module 02 of our project that is Security Threats. Security threats include basic web attacks that could be a cross site scripting or as simple as a click jacking attack. We have depicted two attacks in our module malicious Pop-up and Click Jacking. Let's discuss them in detail.

Malicious Pop-up:

Malicious Pop-up attack means an attack where a third party advertiser advertises their ad having malicious intent behind it. It can be a deceptive or malicious pop-up which randomly appears on the screen of the victim, it usually occurs when using the internet. The pop-up seems legitimate and arousing ultimately manipulates the user into clicking the malicious pop-up. It can be a possibility that you won't see anything happening on your screen and something is being carried out in the back-end forever by clicking malicious pop-up you have allowed the attacker to use your session keys of your banking website session.

The malicious pop-up may look legitimate like click on this link to continue, or allow cookies. The victim may click it considering it to be a valid ad but in fact at backend it allows user to actually authorize attacker of confidential information or access to carry out something on victim's computer.

In our example the malicious pop-up downloads and exe file without the user's consent. That exe files has malicious purpose uploading the files of the user to remote server. Thus, breaching the confidentiality of the user.

Click Jacking:

Click jacking is a type of attack where a legitimate website is covered by an invisible button or a link. Though maybe it would look like you are clicking a legitimate site like a legitimate button but by the intent of it you are actually allowing that button to do changes like steal your cookies, session keys, access to your system's directory etc.

In our case we have Used click Jacking on a login portal and what it does is it installs an extension which ultimately servers as a purpose of a key logger. A key logger is a malicious software to note down whatever the user is typing hence steals the credentials of the user accessing the login portal.

10. Iteration 3

10.1. Use Cases Covered

10.1.1. Extended Use Cases

UC06: USE CASE	Learn Mitigation Techniques
Actors:	End-user
Type:	Primary
Description:	The user enters the application after successful login. The user can see mitigation techniques corresponding to each lab at the end of it.

UC06: USE CASE	Learn Mitigation Techniques
Scope:	The user learns the mitigation techniques.
Level:	User goal level
Primary Actor:	End-user
Stakeholders and interests:	<u>End-user:</u> The user sees mitigation techniques of labs and gets to know how to take necessary steps to avoid cyber-attacks.
Preconditions:	1. The user must log in to their account
Success Guarantee:	The user learns the mitigation techniques.
Main success scenario:	The user learns the litigation techniques.
Extensions:	
Special Requirements:	The user should have an internet connection.
Technology and Data Variations list:	

UC08: USE CASE	Complete lab Topic Content
Actors:	End-user
Type:	Primary
Description:	The user enters the application after successful login. The user access the feature to solve labs and after following the steps the user reaches the end of the lab and completes the lab.

UC08: USE CASE	Complete lab Topic Content
Scope:	The user completes the lab topic.
Level:	User goal level
Primary Actor:	End-user
Stakeholders and interests:	<u>End-user:</u> The user tries to solve labs and after following the steps the user reaches the end of the lab and completes the lab..
Preconditions:	1. The user must log in to their account
Success Guarantee:	The user completes the lab.
Main success scenario:	The user completes the lab topic.
Extensions:	
Special Requirements:	The user should have an internet connection.
Technology and Data Variations list:	

10.1.2. System Sequence Diagrams

System Sequence Diagram 01

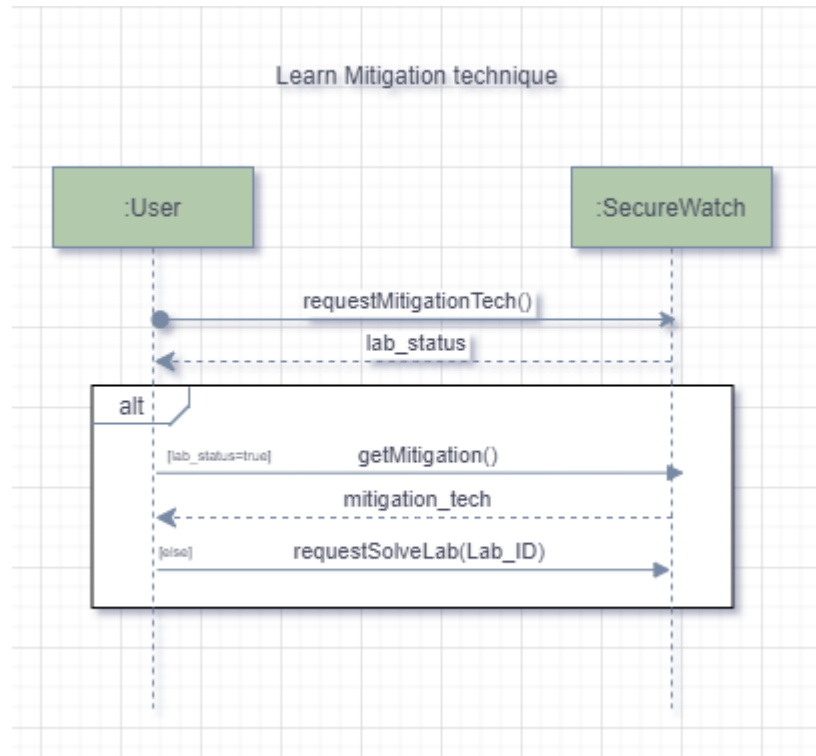
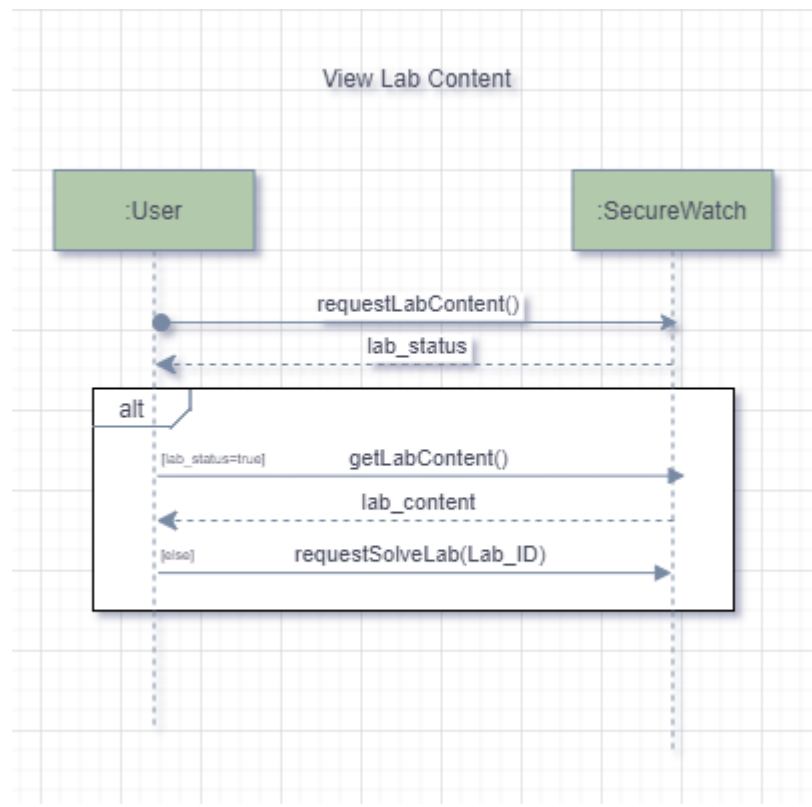
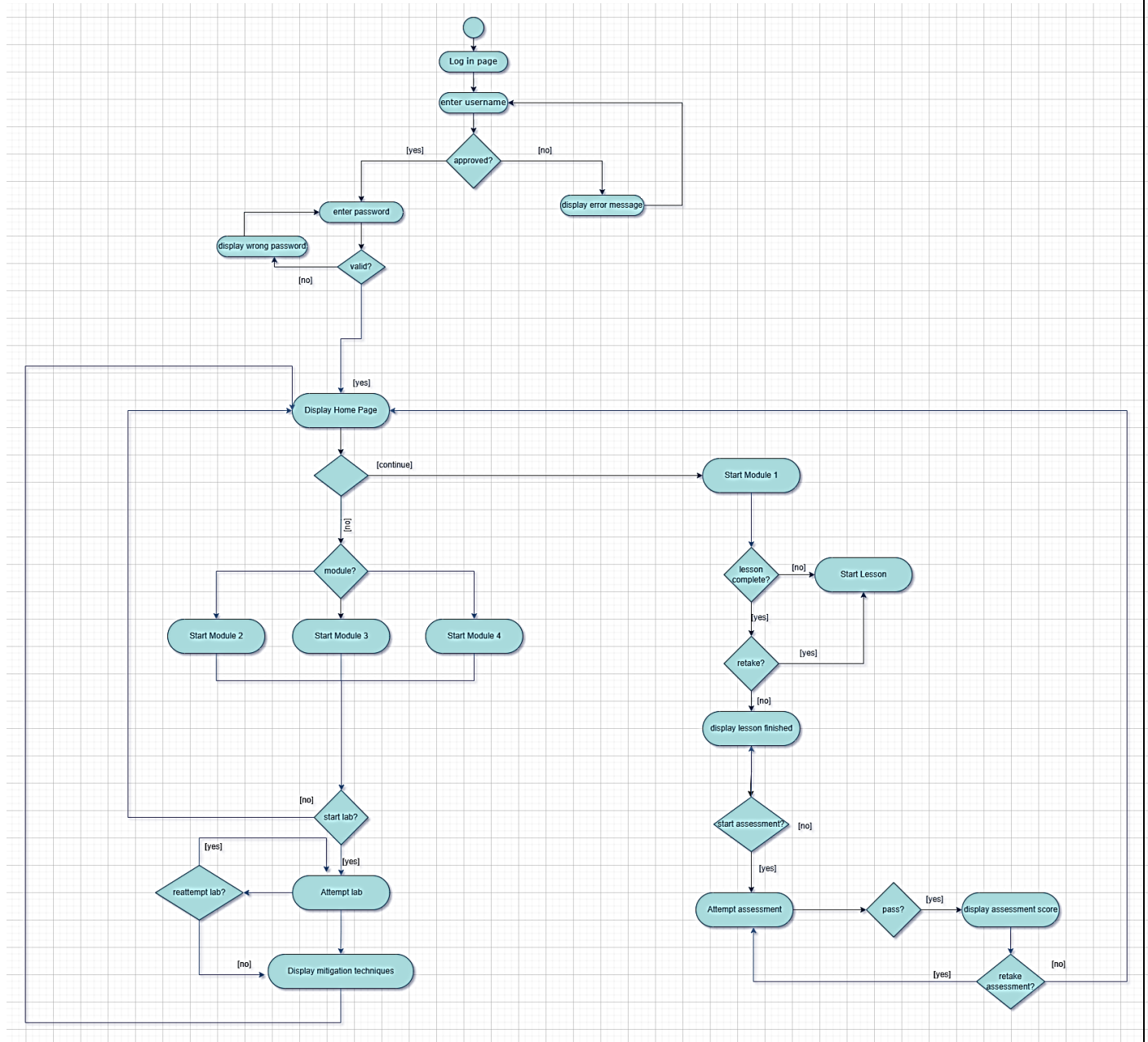


FIGURE 1 MITIGATION TECHNIQUES

System Sequence Diagram 02:**FIGURE 2 LAB CONTENT**

10.1.3. Activity Diagram

After adding all the use cases and their implantation our final activity diagram looks as such



11. Implementation details of Iteration 3

In this Iteration we have covered the lab practices of Module 03 and Module 04 Which included Social Engineering labs and Secure Practices Labs Respectively. We have developed three labs each module.

In third Module namely Social Engineering Module We have made 3 labs, 1 targeting Facebook and 2 targeting Gmail.

Facebook Lab:

The purpose of this Lab is to demonstrate a redirection attack on Facebook. A user normally is scrolling on the internet but a link on legitimate site is using the Turkish Syria donation topic as a bait in to luring the user into going to a site that looks like a legitimate site but is basically a site to get banking credentials of a user. These credentials can be used after the attack to transfer the money from victim's bank account to attacker's bank account.

This is a simple social engineering activity to redirect the user into giving in the confidential information to an attacker who is trying to exploit the trust between user sites and its users.

Gmail Attack Scenario 01:

The motivation behind this attack is to depict a real-life scenario of social engineering attack. For the lab we have created a clone of Gmail. The user receives an email from a trusted source, in this case it was from Outlook asking you to change your password due to certain security issues. The user trusting the sources changes the password. But when the user tries to log back in the password is changed.

In this case the vulnerability is the human, and exploited this vulnerability to get access to the users account.

Gmail Attack Scenario 02:

In the implementation of the Gmail clone lab, the user interface will resemble the Gmail. The user can access their inbox, read and write emails. In this scenario, an employee receives email for boss. The email from the boss shows urgency and asks the employee to immediately send money to the client. The employee without any checks trusts the email and wants to become in the good books of the employer sends the money to the desired client.

Later the employee gets to know that the email was a fraud and he was a victim of the social engineering attack. This scenario is to create awareness among the people to verify even with your closed ones before trusting such emails

In fourth Module we have implemented secure practices such as

HTTPS vs. HTTP

The difference between HTTP and HTTPS connection is shown by the help of an E-commerce website. The attack is demonstrated as such using HTTP website the data is sent over the network in plain text making the sniffing of credentials easier. In contrast, the use of HTTPSS uses SSL certificate to encrypt the data before sending it over the network,

SQL Injection Lab

For this lab we have created a clone of a banking site, where the user logs in. In the first step its is an insecure log in page which is vulnerable to SQL injection attack. When the user enters the list of details from the database become visible.

In the next step, we have created a secure login page which does not allow SQL injection attacks. When the users tries to perform SQL injection he gets no result because proper checks are implemented. The users input is sanitized properly in this case.

We have created a demo SQLite database. Users are pre-added also there is an ‘add account’ section that lets the lab user to add users to the database. Now there’s a login page that let’s user log-in but that is vulnerable it executes Get query that means that fetch data. The problem is that data and SQL statements are attached. We solved this by safe-login page that detaches the data from query using raw Query statement making SQL injection impossible.

Malware Analysis Lab

This lab enables the user to analyze malware both statically and dynamically. The user chooses a file. This file is selected and its hashes are calculated. There is a list of existing malware of multiple files. The selected file hash will be compared to this list of pre-calculated malware hashes. If the hashes are found it indicated the selected file is malware. To dynamically analyze malware, we will be viewing the code of the file and then strings will be checked to detect any malicious activityit comprises.

This lab uses Virus total functions and vendors and casts their votes to determine the integrity and reliability of the file uploaded. It can read contents of file convert file to hash and in the end upload the file the server of virus total. Against that we can get votes of vendors like McAfee and other antivirus solutions and vendors.

12. Iteration 4

In iteration 04 we have integrated our frontend with backend. Also finishing touches of CSS and formatting were added. The use cases were completed in iteration 03 so no more activity and sequence is included.

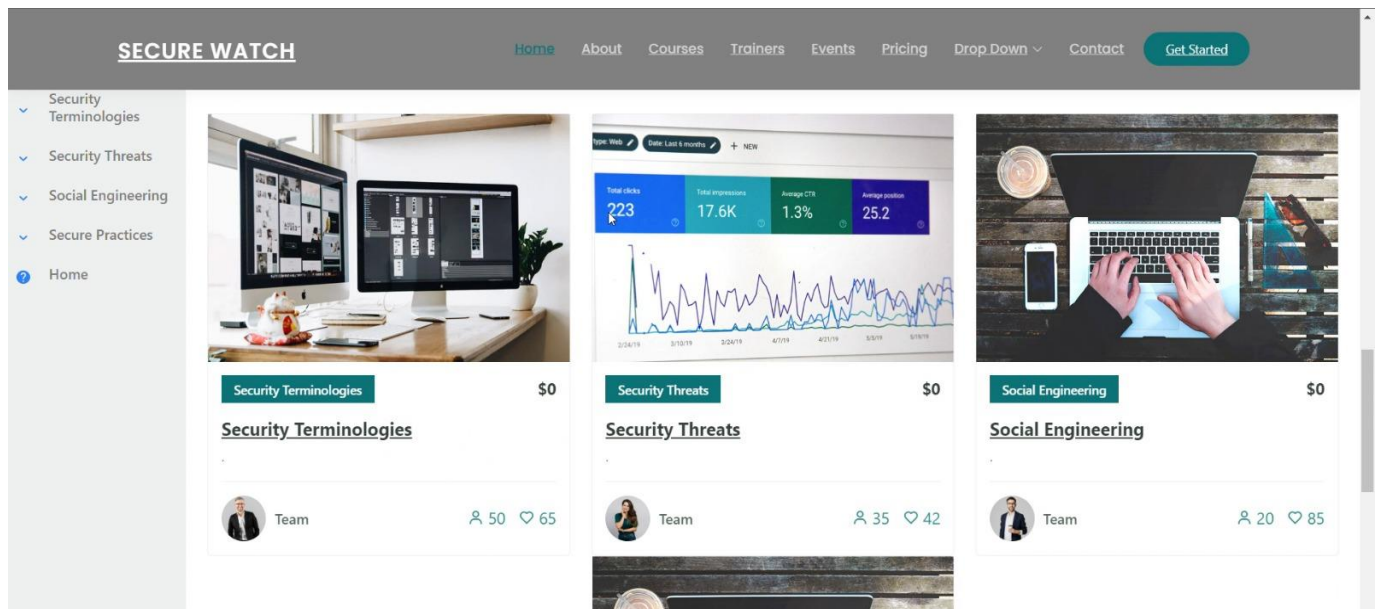
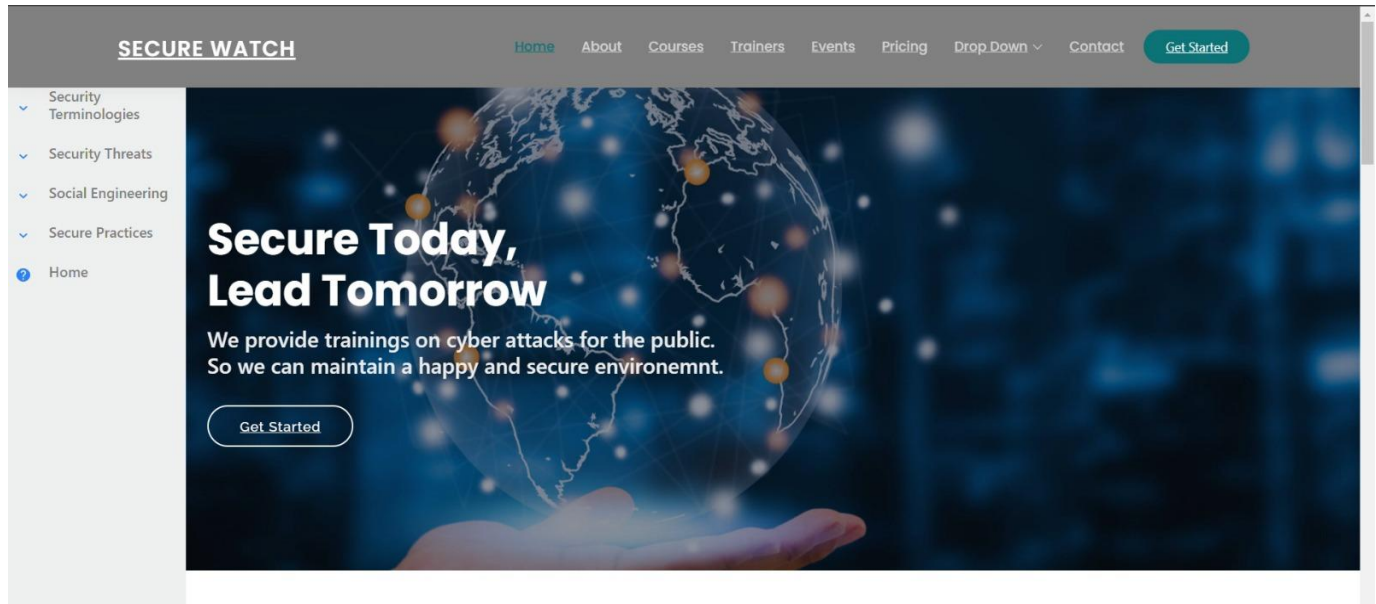
13. Implementation details of Iteration 4

In this iteration we were supposed to integrate our project modules together and after doing so make sure our backend is connected to the front-end activity. The API calling should be ensured to have the exact data printed on the console. The purpose of this iteration is complete the implementation and integrate the parts into one complete working module.

Now after this step the project is ready to present and is in running state to deploy on any other laptop as well. You just have to download and run which will be discussed in the upcoming section.

14. User Manual:

The first step after you Login to the application is that you are faced with the main interface it has navigation bar to easily navigate you to the module. Otherwise, if you scroll down, you'll find the Modules namely Security terminologies, Security Threats, Social Engineering and Secure Practices.



First, we move to Security terminologies and watch the video. Continue to assessment to complete the assessment you can redo or redirect back to main page.

SECURE WATCH

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- Security Terminologies
- Security Threats
- Social Engineering
- Secure Practices
- Home

SECURITY TERMINOLOGIES

A hacker is an individual who uses computers, networking or other skills to overcome a technical problem.

0:10 / 10:33

[Skip to the Assessment](#)

Score:0

Question 1 of 10

In cybersecurity, what does CIA stand for?

- Confidentiality, Integrity, Availability
- Central Intelligence Agency
- Cybersecurity Investigation Agency
- Cybersecurity, Internet, Accessibility

Quit Next

Move on to Next step Click jacking these labs are self-explanatory you follow whatever steps the modal shows you like enter credentials you'll yourself be guided to the next step. Follow the steps as mentioned in the pictures below.

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Security Threats

Social Engineering

Secure Practices

Home

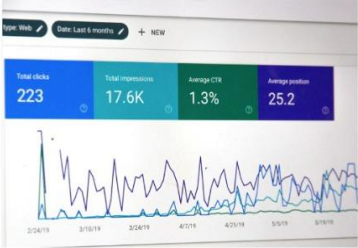
SECURITY THREATS LABS



Clickjacking

\$0

Clickjacking



Malicious Pop-up

\$0

Malicious Pop-up



Register

Username

Email address

Password

Confirm Password

REGISTER

Sign-up Portal.

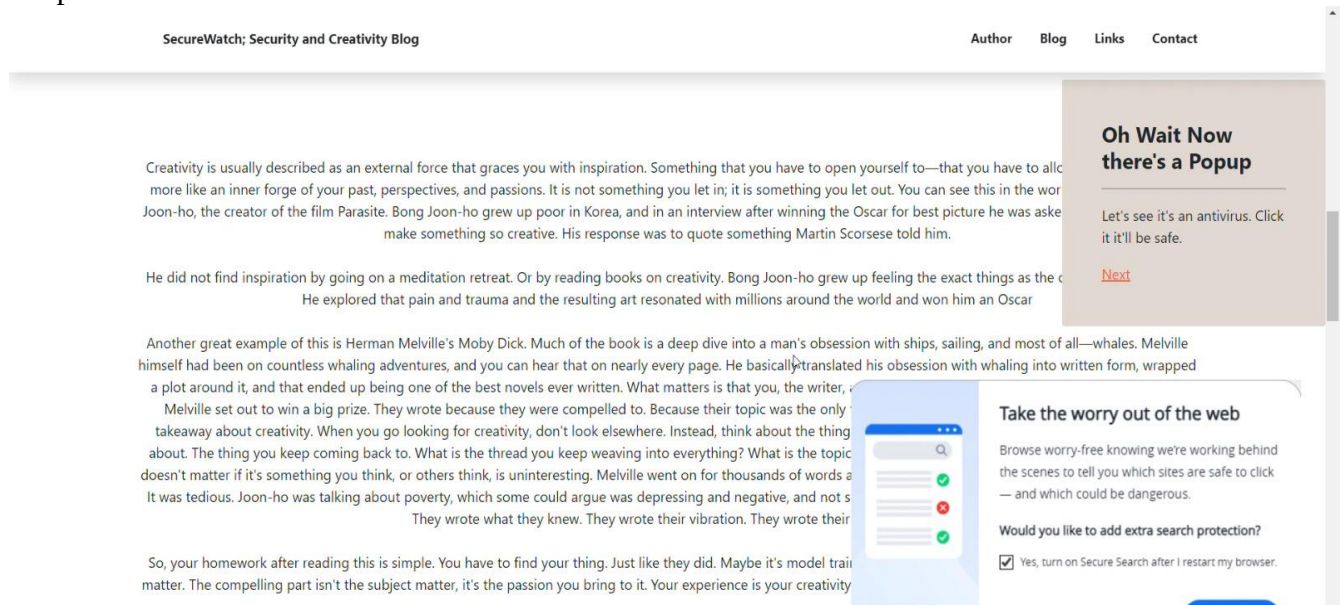
Continue create an account.

Next

Once completed navigate back to main page.

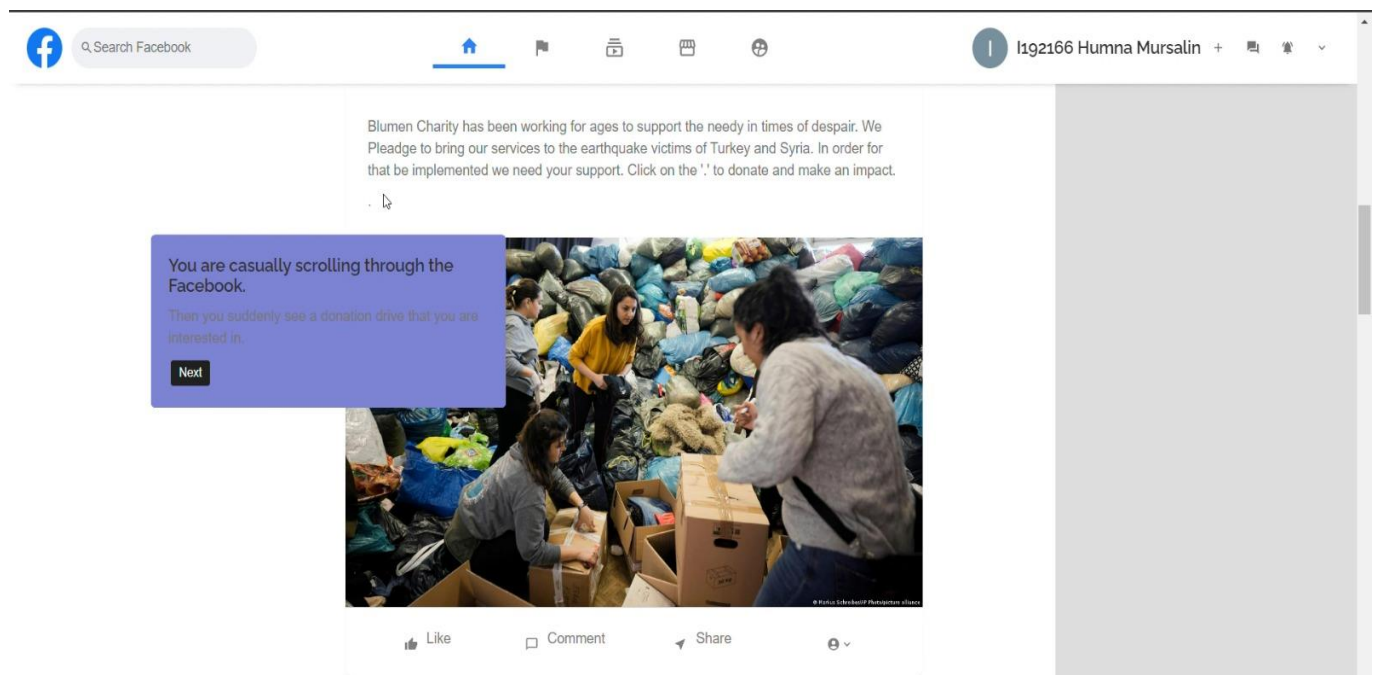
Sunday, 4th June, 2023.

Move on to Next step Malicious Popup these labs are self-explanatory you follow whatever steps the modal shows you like enter credentials you'll yourself be guided to the next step. Follow the steps as mentioned in the pictures below.



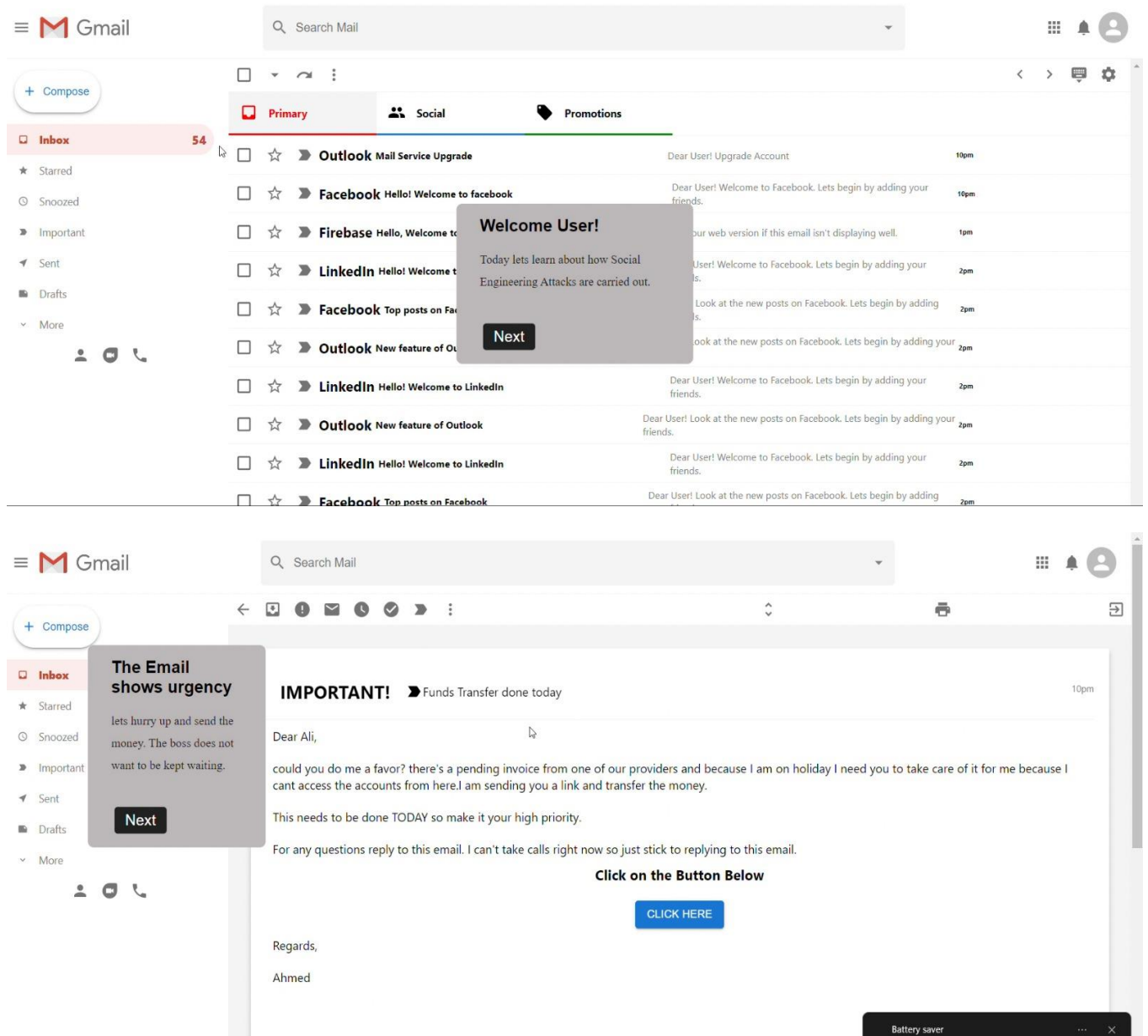
Once completed navigate back to main page.

Move on to Next step Facebook Attack Simulation these labs are self-explanatory you follow whatever steps the modal shows you like enter credentials you'll yourself be guided to the next step. Follow the steps as mentioned in the pictures below.



Once completed navigate back to main page.

Move on to Next step Gmail Attack 01 and 02 these labs are self-explanatory you follow whatever steps the modal shows you like enter credentials you'll yourself be guided to the next step. Follow the steps as mentioned in the pictures below.



Once completed navigate back to main page.

Move on to Next step Malware Analysis these labs are self-explanatory you follow whatever steps the modal shows you like enter credentials you'll yourself be guided to the next step. Follow the steps as mentioned in the pictures below. Check these on virus.exe and installer.exe a malicious and non-malicious file respectively.

The screenshot shows a web application titled "MALWARE ANALYSIS LAB" with a sub-header "MALWARE STATISTIC REPORT GENERATION". The form is divided into three main sections: "Input", "Output", and "Response".

- Input:** Contains a "File Input" section with a "Choose File" button and a text field showing "No file chosen".
- Output:** Contains a text field with the URL "https://www.virustotal.com/api/v3/files/".
- Response:** Contains an empty text field.

At the bottom of the form, there is a "Final" button.

Once completed navigate back to main page.

Move on to Next step SQL Injection these labs are self-explanatory you follow whatever steps the modal shows you like enter credentials you'll yourself be guided to the next step. Follow the steps as mentioned in the pictures below.

The screenshot shows the ALLIANT Bank login page. On the left, there is a "Login" form with fields for "email" (containing "humnamursalin17@gmail.com") and "password" (containing "*****"), and a "Login" button. Below the button is a link: "Don't have an account? Register".

In the center, a green modal overlay displays a list of test credentials:

- First Name:Humna
- PIN:45236
- Bank Balance:12345\$
-
- First Name:Manya
- PIN:Imran
- Bank Balance:12345\$
-
- First Name:Abdullah
- PIN:Akram
- Bank Balance:12345\$
-
- First Name:Ali
- PIN:hussain
- Bank Balance:126785\$

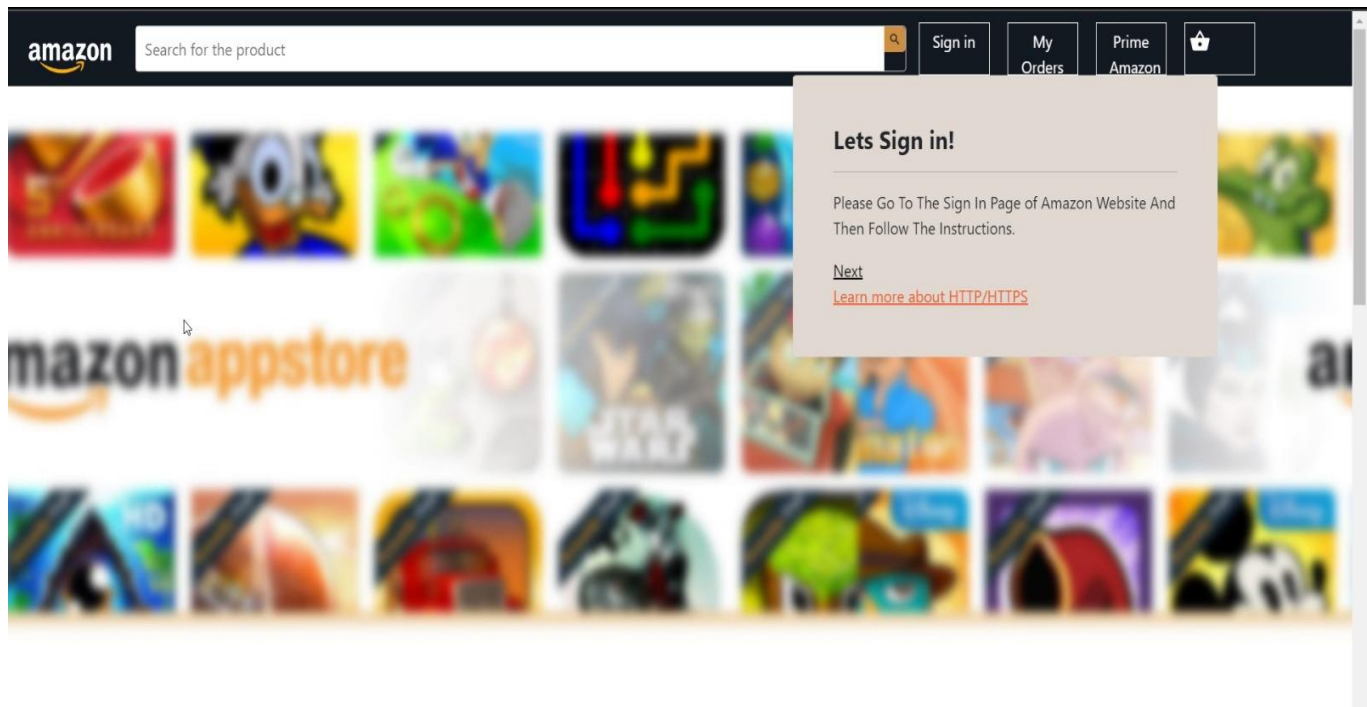
At the bottom of the modal is a "Next" button.

The background shows the ALLIANT Bank website header with navigation links: "Bank", "Bor", "About", and "Become a Member". There is also a search bar and a "Learn more" button on the right.

Once completed navigate back to main page.

Sunday, 4th June, 2023.

Move on to Next step HTTP vs. HTTPS these labs are self-explanatory you follow whatever steps the modal shows you like enter credentials you'll yourself be guided to the next step. Follow the steps as mentioned in the pictures below.

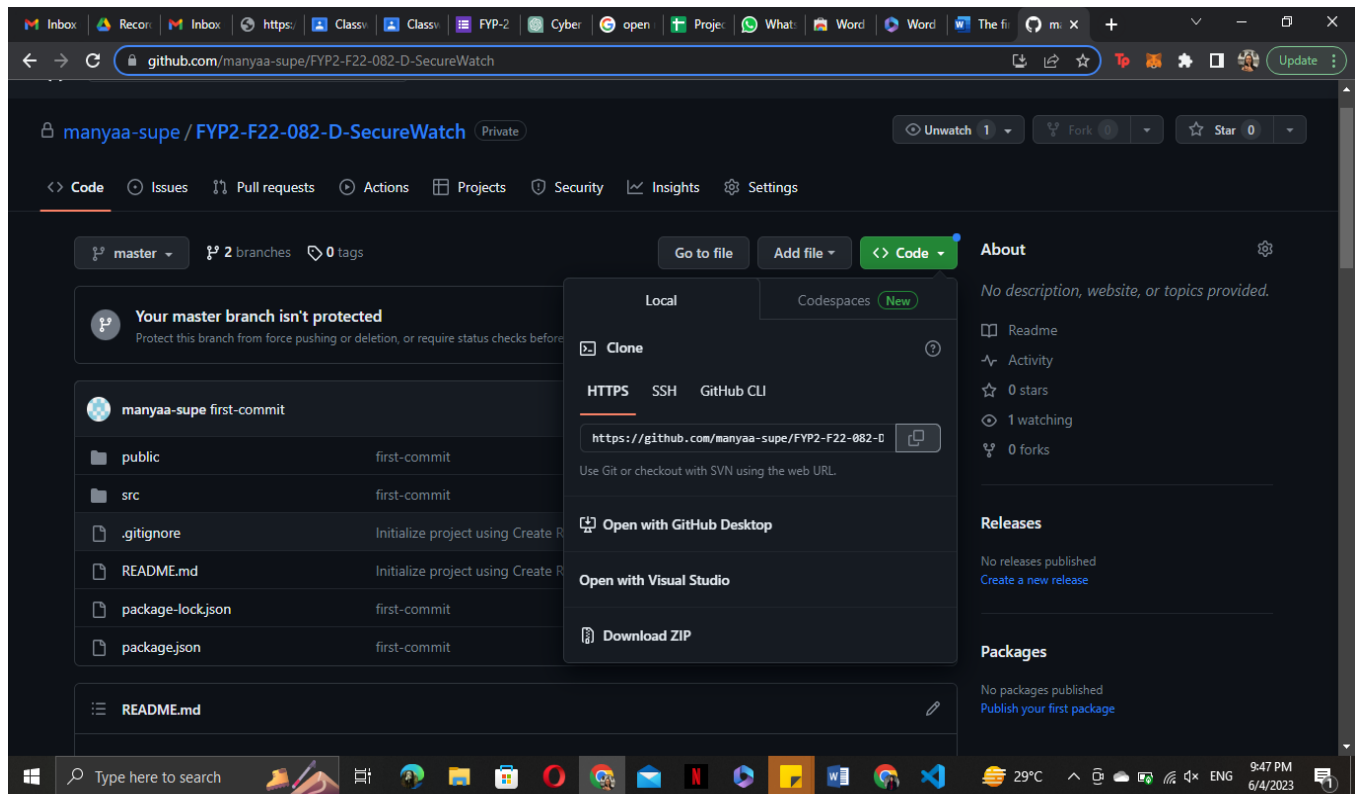


Once completed navigate back to main page.

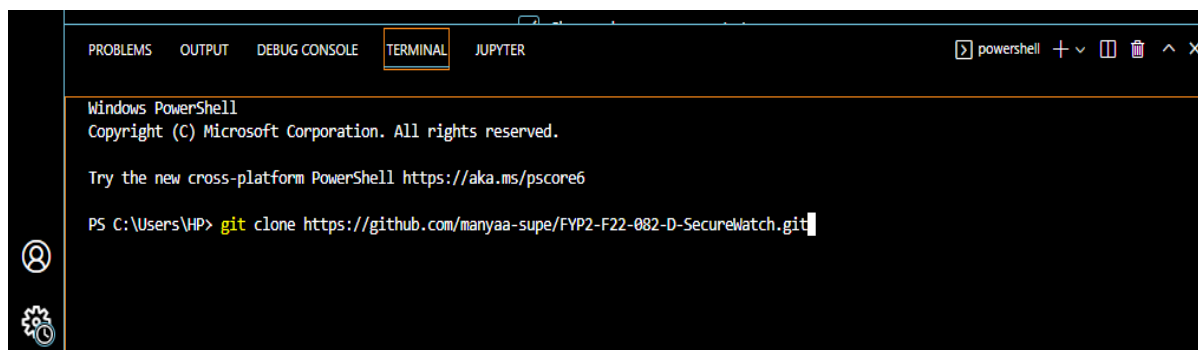
15. Technical Implementation

Go to the provided github link to the project.

Either download project and open in VSCode Or select the following link



Gitclone it in VSCode by opening a new terminal.



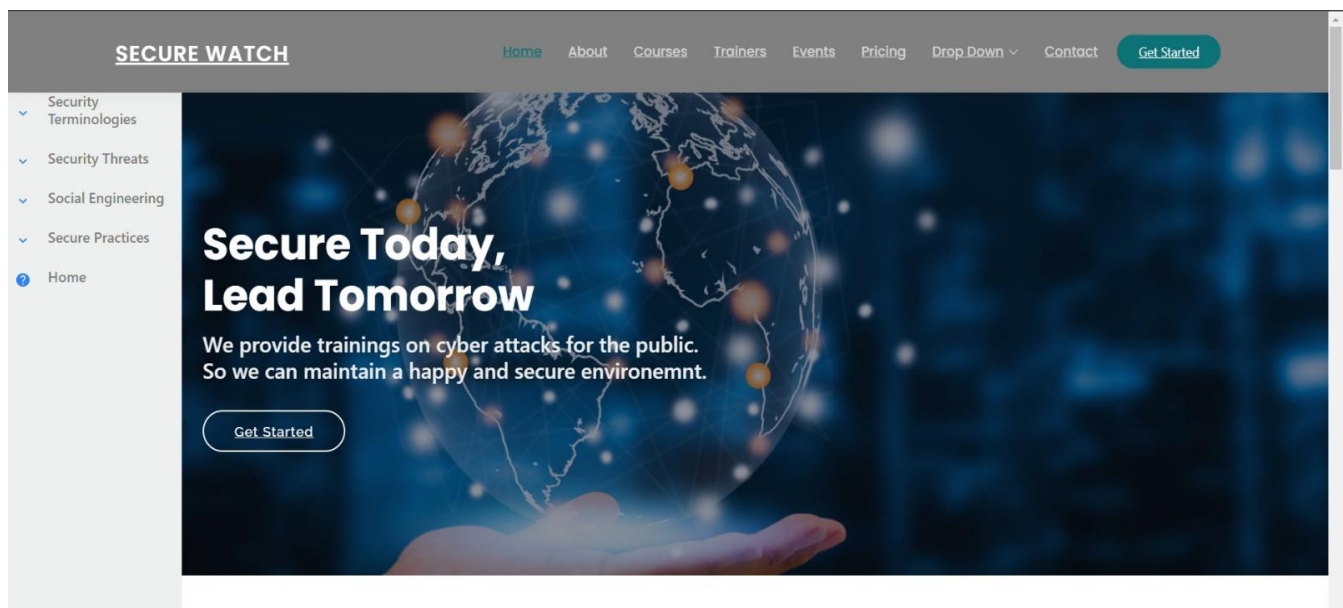
Next cd to server-side to open the server side and run the following commands:

1. `npm i express`
2. `npm i nodemon -D`
3. `npm run dev`

Cd to Client-side and run the following:

1. `npm install`
2. `npm start`

Install additional libraries if not installed. After that you'll see the following Page



16. Testing

A Learning Management system is a platform for providing educational content, test, quizzes, lectures, practical demonstration and in our case practical attack simulation labs. While an LMS can provide all these it typically cannot be tested against some test cases because the test inputs are not usually defined. It is a self-guided platform for teaching of users.

Testing for SecureWatch is not a valid option due to the following reasons:

Learning Management System: It is a learning management system meaning it is a platform to teach cybersecurity skills. It has modules and labs which have a flow all of that flow is pretty self-explanatory.

No dataset is required to test it against that means no input output dataset exists to test the validity of the modules working.

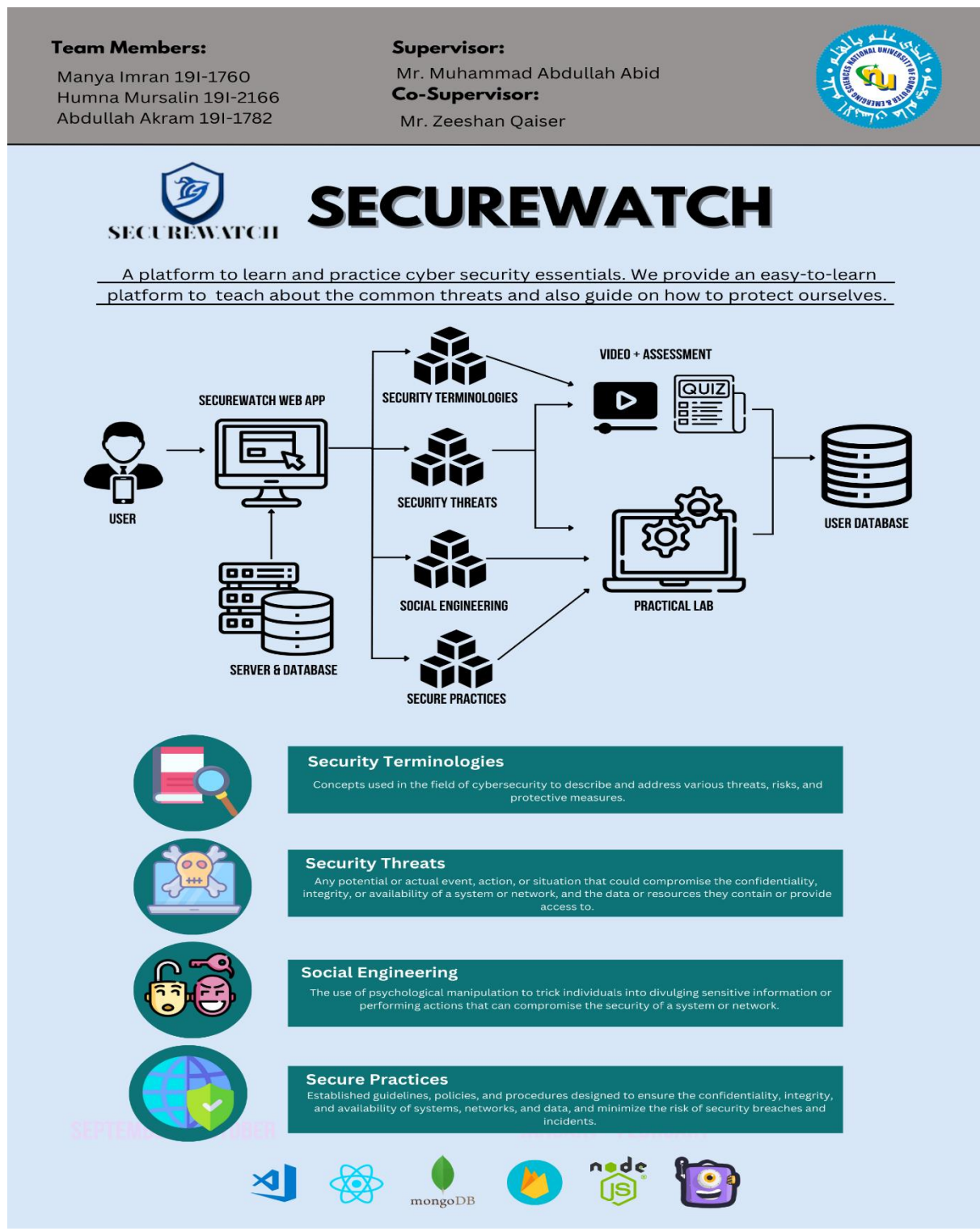
Testing Interface: Test cases require interfaces that require input and against which output can be generated. They put in test input data, execute the project, look at the output, if it matches the output then it is considered correct otherwise it is discarded and test fails. However, in the case of an LMS it provides user friendly interface to provide easy-to-use platform and easy technology to guide and in short it doesn't have data to test it with.

17. Technology Used

We have used Mongo DB database and Firebase database for our data storing. We have used express JS and node JS to add backend functionality and communicate through API calling. Next up, we used React JS to design extremely effective and user-friendly platform for user to get real-world attack experience and to learn techniques.

We have also used Animaker, an animated video creation platform for making content example and definitions explanations.

18. Poster



19. Conclusion

SecureWatch is an innovative and interactive learning management system focused on cyber security. SecureWatch revolutionizes the learning experience by bringing complex cybersecurity concepts to life through animated videos, assessments, and interactive practice labs. Through a series of well-defined virtual labs, we aim to provide learners with real-world experience in tackling cyber-attacks.

SecureWatch is an easy-to-use platform to teach the users of the importance of ensuring cyber security secure practices. It is important to adopt these practices to ensure the security of millions of an organization's customers. It is important to ensure the credibility of the organization and to maintain the trust between the users and the organizations.

SecureWatch is an E-learning platform that is easy to use and super user-friendly. It is not like paid traditional skill-learning platform, the purpose of SecureWatch is to provide free learning that can benefit the entire internet community. SecureWatch also provides lab environments for the users to test their knowledge and skills. SecureWatch is also easy to access anyone can log in and start learning on their devices.

During the designing and development of lessons, we have paid special attention to the cause and results of attacks. It provides updated lectures about the latest news in the cyber security fraud world.

20. **References**

1. <https://www.kaspersky.com/resource-center/preemptive-safety/evil-twin-attacks>
2. <https://www.umass.edu/it/security/phishing-fraudulent-emails-text-messages-phone-calls>
3. <https://www.redteamsecure.com/blog/usb-drop-attacks-the-danger-of-lost-and-found-thumb-drives>
4. <https://www.bitdefender.com/cyberpedia/what-is-online-privacy/>
5. <https://www.crowdstrike.com/cybersecurity-101/malware/malware-analysis/>
6. <https://www.cloudflare.com/learning/ssl/why-use-https/>
7. <https://www.hongkiat.com/blog/five-facebook-attacks/>